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Airlytics: Data-Driven Loyalty Program Optimization for Airline Equity Growth

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Introduction

Today, more than **90% of major global airlines** operate loyalty programs—not as perks, but as **profit engines**. In an era of **rising competition** and thin margins, **loyalty has become a key driver of sustainable growth**, transforming customer data into one of the industry’s most valuable **strategic assets**. Yet many programs still rely on outdated segmentation or aggregate metrics, missing nuanced behavioral signals that could unlock deeper engagement—especially among customers who fly less frequently but hold **high long-term potential**.

This leads to a critical business challenge: **How can data analytics help the airline understand, predict, and influence customer behavior to strengthen loyalty and long-term value?**

The answer lies not just in collecting more data, but in **using it equitably and intelligently**—to uncover hidden patterns, personalize interactions, and ensure that value is shared across *all* customer segments, not just the most visible ones. This means designing systems that balance **commercial goals** with **inclusive growth**.

To address this, we developed a data-driven approach centered on one core mission: **Enhance customer satisfaction, engagement, and loyalty by 20% within the next five years**—not through broad campaigns, but through **precise, insight-led actions** powered by integrated analytics. Our solution combines **Business Intelligence (BI)** to reveal macro trends—such as enrollment gaps or redemption drop-offs—with **Machine Learning (ML)** to predict individual behaviors, like churn risk, tier progression likelihood, or response to personalized offers.

The platform we built processes real-world loyalty data across **six key dimensions**—geography, demographics, flight history, redemption patterns, membership tenure, and engagement frequency—to enable:

1. **Interactive dashboards** that visualize disparities and track KPIs over time,
2. **Predictive models** that forecast customer behaviors—from redemption likelihood and churn risk to tier progression and lifetime value—ensuring proactive, targeted interventions,
3. **Dynamic customer segments** derived from behavioral and demographic clustering, enabling personalized strategies that resonate with diverse traveler profiles.

Unlike conventional systems that optimize only for top-tier frequent flyers or overall revenue averages, our approach balances **multiple strategic goals simultaneously**: reducing churn while increasing redemptions, boosting revenue per customer while expanding participation from underrepresented groups, and accelerating tier progression while personalizing offers based on travel patterns. This holistic approach is achieved through **methodological rigor**—careful handling of class imbalance, validation of multiple models across diverse objectives, and seamless integration of insights into a

unified **web interface** where stakeholders can explore trends, generate predictions, and implement data-driven strategies in real time.

This report documents the complete lifecycle of the Airlytics solution—from building a **star-schema data warehouse** using SSIS and SQL Server, to engineering features, developing and validating predictive models, and deploying an end-to-end application. Every technical decision was guided by a fundamental principle: **data should serve both business objectives and customer needs**. By anchoring analytics to clear strategic goals spanning retention, revenue, engagement, and equity, we demonstrate how modern data engineering can transform airline loyalty programs from transactional systems into engines of **inclusive, sustainable growth**.

Project Summary

Project Summary – Completed Tasks

This section summarizes the distribution of project tasks and details the work contribution by each team member. The efforts are organized across five phases that encompass data analysis, data warehouse design, ETL development, and the implementation of both Business Intelligence dashboards and Machine Learning models.

The project was a collaborative effort among the six team members: Wajd, Khalil, Asma, Adhem, Habiba, and Molka. Initial phases involving data understanding, conceptual design, and infrastructure setup were conducted collectively. In later stages, each member took ownership of a specific business objective, developing tailored analytical solutions and predictive models to address distinct strategic goals.

This table summarizes the completed tasks and identifies the team members responsible for each one.

Task ID	Task Description	Responsible Individual(s)
1	Data and Business Understanding	All members (Wajd, Khalil, Asma, Adhem, Habiba, Molka)
2	Set Objectives Tree	All members (Wajd, Khalil, Asma, Adhem, Habiba, Molka)
3	Data Collection and Storage Area (SA) Setup	All members
4	Data Warehouse (DW) Conceptual Design	All members
5	ETL Process Development	All members
6	Data Warehouse Physical Implementation	All members
7	Visual Studio Connection to SSMS	All members
8	Exploratory Data Analysis (EDA)	All members
9	Data Preparation	All members
10	Feature Selection	Each member separately
11	Business Intelligence (BI) Template Implementation	All members
12	Objective 1: Increase loyalty point redemption by 20% annually over the next 3 years – ML Models Implementation & BI Dashboard Design	Wajd

Task ID	Task Description	Responsible Individual(s)
13	Objective 2: Develop at least three traveler profiles and increase the relevance of personalized offers by 10% within 5 years – ML Models Implementation & BI Dashboard Design	Khalil
14	Objective 3: Increase the registration rate of low-participation regions or socio-economic groups by 10% over 3 years – ML Models Implementation & BI Dashboard Design	Asma
15	Objective 4: Increase customer loyalty by helping 40% of Star members upgrade to Nova or Aurora within 24 months – ML Models Implementation & BI Dashboard Design	Molka
16	Objective 5: Increase Customer Lifetime Value (CLV) by 15% annually over 3 years – ML Models Implementation & BI Dashboard Design	Habiba
17	Objective 6: Reduce the current churn rate by 10% within 18 months – ML Models Implementation & BI Dashboard Design	Adhem
18	Website Template Development	All members
19	Embedded Power BI Dashboards	All members
20	Machine Learning Models Integration	All members
21	Client Single Form (Run All Models for User-Friendly Application)	All members
22	Role-Based Access Implementation	All members
23	Chatbot for Decision-Making Support	All members

Team Members: Wajd, Khalil, Asma, Adhem, Habiba, Molka

Chapter 1

Context and Business Understanding

Introduction

This chapter establishes the business context of the Airlytics project by examining the strategic role of loyalty programs within the airline industry. It highlights how these programs have become central to revenue generation and customer relationship management, while also facing growing challenges related to engagement, retention, and inclusivity.

The chapter then introduces Airlytics as a data-driven analytics platform designed to address these challenges by transforming loyalty data into actionable insights. Finally, it outlines the strategic objectives that guide the platform's development and define its expected business impact.

1.1 Airline Industry and Loyalty Programs Context

Airline loyalty programs have evolved from simple frequent flyer schemes into major revenue sources. Today, these programs generate significant income through co-branded credit cards, commercial partnerships, and reward redemptions, reaching up to **40% of total airline revenue** for some carriers. They have become sophisticated marketing ecosystems that extend far beyond aviation to include hotel partnerships, retail collaborations, and digital payment systems.

As shown in Figure 1.1, major loyalty programs generated between \$1 billion and \$6.5 billion in 2023. For instance, Southwest's Rapid Rewards represents 22% of total revenue, while Delta's SkyMiles and American's AAdvantage each contribute over 12%, confirming their role as critical profit engines.

Given this financial importance, modern loyalty programs must balance two key objectives:

- **Maximize revenue:** Offer personalized rewards and partnerships to boost spending and program profitability.
- **Ensure inclusivity:** Provide fair access to loyalty benefits across regions and customer groups, avoiding inequities

Achieving these objectives requires data-driven optimization that identifies both high-value opportunities and participation barriers, transforming loyalty programs into proactive engagement platforms.

Multi-Billion Dollar Club of FFPs - 2023			
Results may include sales of miles to partners and service fees charged to members			
Airline	Program Name	FFP Revenue	As a % of Airline Revenue
Delta	SkyMiles	\$6,555,000,000	12.0%
American	AAdvantage	\$6,527,000,000	12.4%
Southwest	Rapid Rewards	\$5,839,000,000	22.4%
United	MileagePlus	\$5,529,000,000	10.3%
Alaska Air Group	Mileage Plan	\$1,682,000,000	16.1%
Qantas Group	Frequent Flyer	\$1,459,000,000	14.1%
Air Canada	Aeroplan	\$1,267,000,000	7.7%
Int'l Airlines Group	Avios	\$1,126,000,000	3.5%
JetBlue	TrueBlue	\$1,027,000,000	10.7%
Source: Financial disclosures made by airlines in annual reports. Qantas fiscal year ends 30 June. Currency conversion using 01 July 2023 rate at XE.com.			

Figure 1.1: Multi-Billion Dollar Club of Frequent Flyer Programs - 2023

1.2 Challenges in Loyalty Programs

Airline loyalty programs face several key challenges that limit their effectiveness and inclusivity:

- **Unequal participation:** Enrollment varies widely by region and demographic. Some areas or income levels have very low signup rates, even though those customers fly frequently. This gap often stems from factors like program complexity or lack of awareness, representing both missed revenue and equity concerns.
- **Low engagement and high churn:** Many members earn points but never redeem them. A large portion of loyalty points go unused each year. Additionally, many mid-level members gradually stop using the program (churn), and airlines often struggle to identify these at-risk customers in time.
- **Untapped customer value:** Airlines often focus rewards on their already most loyal customers but miss opportunities to grow value with others. Without good analytics, it's hard to know which casual or newer members could become high-value customers with the right incentives.
- **Lack of actionable insights:** Traditional reporting can show *what* is happening (for example, falling redemption rates) but not *why*. Without predictive analytics and clear recommendations, airlines find it difficult to turn data into concrete actions—such as knowing which customers to target for retention.

These challenges highlight the need for smarter, data-driven tools to make loyalty programs both more efficient and more equitable.

1.3 The Airlytics Platform: Data-Driven Solutions

Airlytics is a comprehensive analytics platform designed to turn loyalty data into actionable intelligence that addresses these challenges. It consolidates millions of customer and flight records in a unified data warehouse and provides interactive tools for visualization and machine learning.

Key capabilities include:

- **Interactive dashboards:** Visual reports show real-time and historical trends in program enrollment, point redemptions, churn rates, and revenue by different customer segments (regions, income levels, tiers, etc.). These dashboards help spot patterns and disparities at a glance.
- **Predictive analytics:** Machine learning models forecast individual customer behaviors. For example, they can predict which members are likely to stop using the program, how likely someone is to redeem points, or estimate each customer's lifetime value. This enables proactive engagement with the right people at the right time.
- **Customer segmentation:** The platform automatically groups travelers by their travel and spending habits. Identifying clusters such as "frequent short-haul flyers" or "infrequent but high-value travelers" allows for personalized marketing strategies for each group.
- **Actionable recommendations:** Based on the predictions and segments, Airlytics suggests specific actions. For example, it might identify a set of high-churn customers to target with special offers, or a group of under-utilizing members to encourage point redemptions.
- **User-friendly interface:** The system is accessible via an easy-to-use web portal exclusively for airline team members. Staff can explore the data, apply filters (by time period, location, demographics, etc.), and run on-demand predictions for individual customers or groups—without needing technical expertise. Access is role-based, ensuring each team member sees only information relevant to their responsibilities.

To ensure effective adoption, the platform meets the following operational requirements:

- **Easy to use:** Intuitive interface designed for business users with minimal training required.
- **Fast performance:** Dashboards load in under 3 seconds; predictions generate in under 5 seconds for individuals or 30 seconds for batches of 1,000 customers.
- **Accurate and current:** Daily data updates ensure predictions reflect the latest information, with models maintaining high accuracy standards (80% recall for churn prediction, 0.6 silhouette score for clustering).
- **Scalable:** Supports multiple simultaneous users and can accommodate growth to 10+ million customer records.
- **Secure:** Encrypted data with role-based access controls ensuring team members only see relevant information. Full audit logging tracks all activities for accountability.
- **Maintainable:** Models retrained quarterly with updated data, supporting easy updates and rollback capabilities.

In summary, Airlytics transforms raw loyalty data into insights and recommendations that drive both revenue growth and more inclusive program participation.

1.4 Strategic Objectives and Impact

Airlytics directly supports the airline's strategic goals by linking analytics to business objectives. The overarching mission is to **boost overall customer satisfaction, engagement, and loyalty by 20% over five years**. This mission branches into six interconnected objectives, illustrated in Figure 1.2, where each goal reinforces the others to create a comprehensive growth strategy.

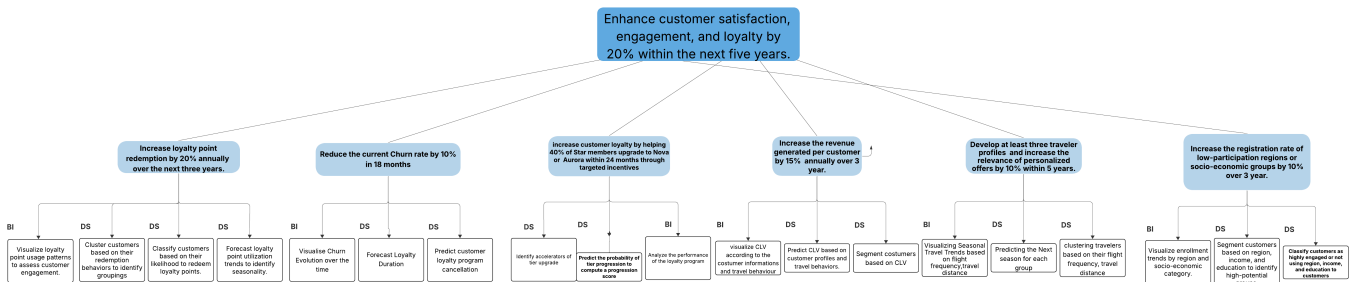


Figure 1.2: Strategic Objectives Hierarchy for Airlytics Platform

1.4.1 The Six Strategic Objectives

Each objective is supported by a combination of data visualization for monitoring progress and predictive analytics for guiding actions:

- 1. Boost point redemption by 20% annually:** Increase the percentage of earned loyalty points that members actually use. Data clustering identifies customer groups with similar redemption patterns, while prediction models determine who is most likely to redeem points with the right incentives. Dashboards track redemption trends across seasons and customer segments.
- 2. Reduce churn by 10% in 18 months:** Lower the rate at which members drop out of the program. By predicting loyalty program cancellations and forecasting membership duration, the airline can intervene early with at-risk customers. Visualizations show churn evolution over time, helping identify which customer groups need attention.
- 3. Encourage tier upgrades (40% of Star to Nova/Aurora in 24 months):** Help members move to higher loyalty tiers, as higher-tier members typically spend more. By understanding which factors accelerate tier progression—such as frequent bookings or premium service usage—the airline can motivate mid-tier members with tailored offers and targeted incentives.
- 4. Increase revenue per customer by 15% annually:** Grow the average value each customer brings over their lifetime. Predicting customer lifetime value (CLV) and segmenting customers by potential value allows the airline to focus marketing efforts and rewards on the most promising segments, maximizing return on investment.
- 5. Personalize offers (develop 3+ traveler profiles, +10% relevance in 5 years):** Create distinct traveler profiles such as "frequent business flyers," "occasional holiday travelers," or "weekend adventurers." Better personalization—powered by clustering and seasonal prediction—leads to offers that customers are more likely to respond to, increasing both satisfaction and conversion rates.

6. **Expand inclusivity (increase low-region enrollment by 10% in 3 years):** Raise enrollment and participation in underserved regions or demographic groups. By identifying high-potential but low-participation segments through demographic analysis and engagement prediction, the airline can adjust its outreach and program design to be more inclusive and equitable.

Table 1.1: How Analytics Support Each Objective

Objective	Predictions & Segmentation (DS)	Dashboards & Tracking (BI)
1. Boost redemptions	- Group customers by redemption behavior - Predict redemption likelihood - Forecast seasonal redemption trends	Track point usage patterns and engagement levels
2. Reduce churn	- Predict membership duration - Identify cancellation risk	Visualize churn rates over time by segment
3. Encourage upgrades	- Identify upgrade accelerators - Calculate tier progression scores	Monitor tier distribution and progression rates
4. Increase revenue	- Predict customer lifetime value - Segment by value potential	Display CLV across customer segments
5. Personalize offers	- Predict next travel season - Cluster by travel patterns	Show seasonal trends by traveler type
6. Expand inclusivity	- Segment by region and demographics - Predict engagement likelihood	Track enrollment gaps across regions

1.4.2 The Connected Impact

These objectives operate as a unified system: reducing churn protects revenue gains from increased redemptions, tier upgrades enhance customer lifetime value, and expanded inclusivity grows the base of potential high-value customers. By aligning analytical insights with strategic goals, Airlytics ensures that progress in one area reinforces the others.

To achieve these strategic objectives and ensure their operational feasibility, Airlytics relies on a robust and well-structured technical foundation. The following section presents the technical stack that supports data processing, analytics, visualization, and platform deployment.

1.5 Technical Stack

The Airlytics platform is built on a modular technical stack organized into three complementary layers: data engineering, machine learning, and business intelligence with web application. This layered architecture ensures scalability, maintainability, and a clear separation of responsibilities across the system.

1.5.1 Data Engineering Layer

The data engineering layer is responsible for data ingestion, transformation, and storage, forming the foundation of the analytical platform.

- **Microsoft Excel** was used during the early stages of the project for initial data inspection, validation, and exploratory analysis on limited data samples.



Figure 1.3: Microsoft Excel Logo

- **SQL Server Integration Services (SSIS)** was employed to implement automated ETL pipelines, enabling the extraction, transformation, and loading of customer, flight, and loyalty data into the data warehouse.
- **SQL Server**, managed through **SQL Server Management Studio (SSMS)**, serves as the central data warehouse. It stores structured data organized in a star schema and supports analytical querying and feature preparation for downstream tasks.



Figure 1.4: SQL Server Logo

1.5.2 Machine Learning Layer

The machine learning layer focuses on data analysis, experimentation, and predictive model development.

- **Python** is used as the primary programming language for data analysis and machine learning tasks.



Figure 1.5: Python Logo

- **Jupyter Notebook** provides an interactive environment for exploratory data analysis, feature engineering, and model experimentation.



Figure 1.6: Jupyter Notebook Logo

- **Google Colab** is used to execute notebooks in a cloud-based environment, enabling faster experimentation and access to scalable computational resources.



Figure 1.7: Google Colab Logo

1.5.3 Business Intelligence and Web Application Layer

This layer delivers analytical results to end users through dashboards and a web-based interface.

- **Power BI** is used to design interactive dashboards that monitor key performance indicators related to loyalty program objectives.



Figure 1.8: Power BI Logo

- **Django** serves as the web application framework, integrating dashboards, analytical outputs, and user interactions within a unified platform.



Figure 1.9: Django Logo

- **HTML5, CSS3, and JavaScript** are used to build a responsive and user-friendly frontend interface.



Figure 1.10: Frontend Technologies Logo: HTML5, CSS3, and JavaScript

- **Visual Studio Code** is used as the primary development environment for backend and frontend code, supporting efficient development and debugging



Figure 1.11: Visual Studio Code Logo

- **SQLite** supports lightweight storage of application-level data such as user accounts and prediction logs.

1.5.4 Report and Documentation Layer

This layer supports project documentation and report production.

- **Overleaf** is used as a collaborative $\text{\LaTeX}$ environment for writing, editing, and managing the project report and technical documentation.



Figure 1.12: Overleaf Logo

Conclusion

This chapter presented the strategic foundations and technical underpinnings of the Airlytics platform. It first analyzed the business importance of airline loyalty programs and identified the key challenges that limit their effectiveness, including uneven participation, customer churn, and under-utilized customer value.

In response to these challenges, Airlytics was introduced as an analytics-driven solution aligned with clearly defined strategic objectives focused on engagement, revenue growth, personalization, and inclusivity. The chapter then detailed the technical stack supporting the platform, illustrating how data engineering, machine learning, business intelligence, and documentation tools work together to deliver a scalable and actionable analytics system.

Together, these elements establish a solid foundation for the analytical models and results presented in the following chapters.

Chapter 2

Data Integration

Introduction

This chapter presents the data integration and preparation phase of the Airlytics platform, which forms the foundation for all subsequent analytical activities. Effective analytics require high-quality, well-structured data organized in a way that supports both operational efficiency and analytical flexibility.

The chapter begins by describing the source data systems and their characteristics. It then details the construction of a staging area (SA_airline) where raw data is consolidated and prepared. Next, it presents the design and implementation of the data warehouse (Airline_DW) using a star schema architecture optimized for analytical queries. The chapter concludes by examining the ETL processes that populate these systems and the data quality measures implemented throughout.

2.1 Source Data Systems

The Airlytics platform integrates data from four CSV files containing comprehensive information about 160,000 customers and their interactions with the airline's loyalty program. Understanding these source files is essential for designing effective data integration processes.

2.1.1 Source Data Files

The airline's operational data originates from four primary CSV files:

- **Customer Loyalty History (customer_loyalty_history.csv):** Contains the complete history of loyalty program participation including loyalty numbers, enrollment dates, membership tiers (Star, Nova, Aurora), enrollment types, tier changes, points accumulated and redeemed, and cancellation events. This file tracks the lifecycle of each customer's engagement with the loyalty program over time.
- **Customer Flight Activity (customer_flight_activity.csv):** Maintains comprehensive flight records including customer identifiers, booking information, travel dates, flight codes, routes, service classes (Economy, Business, First), travel types, flight distances, ticket costs, and service purchases. This file captures the complete transactional history of customer travel behavior, including demographics such as gender, education level, salary, and marital status.

- **Calendar (calendar.csv):** Provides a comprehensive date dimension covering the full analytical period with hierarchical time attributes including years, semesters, quarters, and months. This file supports time-based analysis and temporal aggregations.
- **Customer Satisfaction Survey (satisfaction_survey.csv):** Contains post-flight satisfaction ratings collected after customer journeys. Each survey is identified by a unique feedback code and includes overall satisfaction scores on a defined scale, enabling customer experience analysis and service quality assessment.

2.1.2 Data Volume and Characteristics

The source dataset presents the following characteristics:

- **Customer base:** 160,000 unique customers with varying levels of loyalty program engagement, from newly enrolled members to long-term participants
- **Data format:** All source files are provided in CSV (Comma-Separated Values) format, requiring parsing and type conversion during the loading process
- **Temporal coverage:** Data spans multiple years of operational history, enabling longitudinal analysis of customer behavior and program evolution
- **Data relationships:** Customer records link to multiple flight activities, loyalty transactions, satisfaction surveys, and geographic locations through common identifiers

2.1.3 Data Quality Challenges

Initial profiling of the source CSV files revealed several data quality issues requiring attention:

- **Missing values:** Optional fields such as cancellation dates and certain demographic attributes contain null values that must be handled appropriately
- **Format inconsistencies:** Date formats vary across files; text fields exhibit inconsistent capitalization and spacing
- **Data type mismatches:** Numeric fields stored as text require conversion; date strings need parsing into proper date types
- **Duplicate records:** Some data identifiers appear multiple times, requiring deduplication logic based on business rules
- **Referential integrity gaps:** Not all customer IDs in flight activity have corresponding entries in loyalty history, requiring orphan record handling

These characteristics necessitated a staged approach to data integration, beginning with a staging area for data consolidation and cleansing before loading into the analytical data warehouse.

2.2 Staging Area Design and Implementation

The staging area (SA_airline) serves as an intermediate data repository where source CSV files are loaded, validated, and prepared for transformation into the data warehouse structure. This layer provides a buffer between raw source data and the analytical environment, allowing for data quality checks and transformations without repeatedly accessing the original files.

2.2.1 Staging Area Architecture

The staging area was implemented in SQL Server and organized around the four source CSV files:

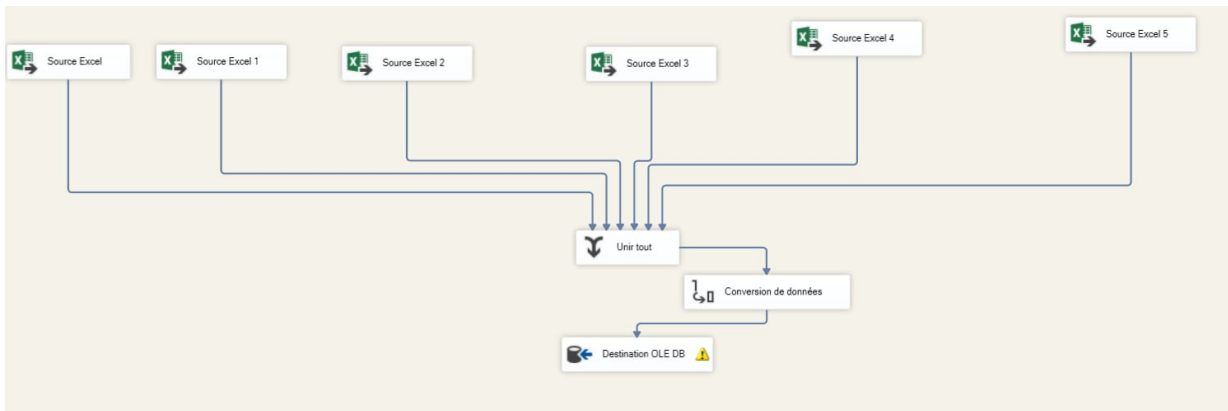


Figure 2.1: Staging Area Database Structure (SA_airline)

Key staging tables include:

- **dbo.Customer_Flight_Activity:** Direct load of the customer flight activity CSV file, preserving all original columns including customer demographics, flight details, service classes, travel types, distances, costs, and service purchases. This table serves as the primary source for customer attributes and transactional behavior.
- **dbo.Customer_Loyalty_History:** Stores the complete loyalty program participation history including enrollment information, membership tiers, points transactions, and cancellation records loaded from the customer loyalty history CSV file.
- **dbo.Calendar:** Contains the date dimension data loaded from the calendar CSV file, providing temporal attributes for time-based analysis.
- **dbo.Customer_Satisfaction_Survey:** Holds customer feedback data with satisfaction ratings and feedback codes loaded from the satisfaction survey CSV file.

Additionally, the staging area includes utility tables:

- **dbo.fact_customer_sa:** A pre-aggregated staging table that consolidates customer metrics from the source files before final loading into the data warehouse fact table
- **Graph Tables:** Visual representations of table relationships used for validation and documentation purposes

2.2.2 CSV Loading Process

Data flows from CSV files into the staging area through SSIS packages designed specifically for flat file processing:

1. **File validation:** Check file existence, accessibility, and basic format consistency (delimiter verification, header row validation)
2. **Flat file connection:** Configure SSIS flat file connection managers with appropriate code page (UTF-8), column delimiters, and text qualifiers
3. **Row-by-row loading:** Process CSV rows sequentially and handling conversion errors gracefully
4. **Error row handling:** Rows that fail conversion are redirected to error tables with full row content and error descriptions preserved for investigation
5. **Load verification:** Compare loaded row counts against source CSV file row counts; validate that all expected columns were loaded

2.2.3 Data Quality Controls in Staging

To ensure data quality in the staging area, several controls were implemented:

Data profiling reports generated after staging load provide statistics on:

- Total records loaded per file
- Min/max/average for numeric fields
- Records failing validation rules

2.3 Data Warehouse Design

The Airline_DW data warehouse implements a dimensional modeling approach based on the star schema, optimized for analytical query performance and business user comprehension. This design separates descriptive attributes into dimension tables and measurable facts into a central fact table.

2.3.1 Star Schema Architecture

The data warehouse follows a star schema pattern with one central fact table surrounded by seven dimension tables:

This architecture provides several advantages:

- **Query performance:** Simplified join paths enable fast query execution for dashboards and reports
- **Business clarity:** Intuitive structure aligns with business concepts (customers, flights, time periods, geography)
- **Flexibility:** New dimensions can be added without restructuring existing tables
- **Aggregation efficiency:** Dimensional attributes support efficient grouping and filtering operations

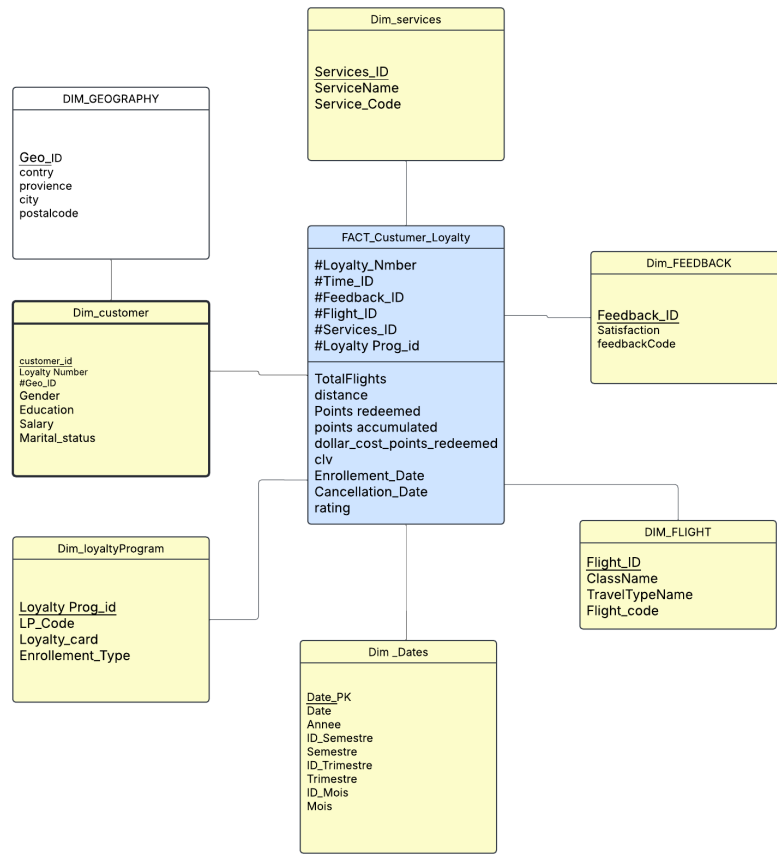


Figure 2.2: Airline_DW Star Schema Architecture

2.3.2 Dimension Tables

Each dimension table contains descriptive attributes that provide context for analyzing facts. Dimensions use business keys from the source systems as primary keys, maintaining referential relationships with the fact table.

2.3.2.1 dim_customer

The customer dimension captures demographic attributes and loyalty identifiers:

Table 2.1: Customer Dimension Structure

Field Name	Data Type	Description
customer_id	VARCHAR(50) (PK)	Unique customer identifier
loyalty_number	VARCHAR(50)	Loyalty program membership number
geo_id	VARCHAR(20)	Foreign key to geography dimension
gender	VARCHAR(10)	Customer gender (Male/Female)
education	VARCHAR(50)	Highest education level attained
salary	VARCHAR(30)	Salary range or income bracket
marital_status	VARCHAR(20)	Marital status (Single/Married/Divorced/Widowed)

This dimension supports demographic segmentation for targeted marketing and customer profiling. The geo_id field links to detailed geographic information in the geography dimension.

2.3.2.2 dim_dates

The date dimension provides comprehensive temporal attributes with hierarchical time structure:

Table 2.2: Date Dimension Structure

Field Name	Data Type	Description
date_pk	INT (PK)	Primary key (YYYYMMDD format)
date	DATE	Actual calendar date
annee	INT	Calendar year (année)
id_semestre	INT	Semester identifier (1-2)
semestre	VARCHAR(20)	Semester name
id_trimestre	INT	Quarter identifier (1-4)
trimestre	VARCHAR(20)	Quarter name
id_mois	INT	Month number (1-12)
mois	VARCHAR(20)	Month name

This dimension enables temporal aggregations at multiple levels (yearly trends, semester comparisons, quarterly analysis, monthly patterns) and supports business calendar logic. The French nomenclature (année, semestre, trimestre, mois) reflects the international context of the airline operations.

2.3.2.3 dim_feedback

The feedback dimension captures customer satisfaction survey information:

Table 2.3: Feedback Dimension Structure

Field Name	Data Type	Description
feedback_id	INT (PK)	Unique feedback identifier
feedback_code	VARCHAR(50)	Business code for feedback record
satisfaction	INT	Overall satisfaction rating score

This dimension supports customer experience analysis and enables correlation between satisfaction levels and other customer behaviors. The satisfaction score represents aggregated feedback from post-flight surveys.

2.3.2.4 dim_flight

The flight dimension describes service and travel characteristics:

Table 2.4: Flight Dimension Structure

Field Name	Data Type	Description
flight_id	INT (PK)	Unique flight identifier
class_name	VARCHAR(30)	Service class (Economy, Business, First Class)
travel_type_name	VARCHAR(50)	Type of travel (Domestic, International, Regional)
flight_code	VARCHAR(20)	Business code for flight record

This dimension supports analysis of service class preferences, travel pattern segmentation, and route type profitability. The travel_type_name enables differentiation between domestic and international customer behaviors.

2.3.2.5 dim_geography

The geography dimension provides hierarchical location information:

Table 2.5: Geography Dimension Structure

Field Name	Data Type	Description
geo_id	VARCHAR(20) (PK)	Unique geographic identifier
country	VARCHAR(50)	Country name
province	VARCHAR(50)	Province or state name
city	VARCHAR(50)	City name
postal_code	VARCHAR(20)	Postal or ZIP code

The geography dimension enables geographic segmentation for market analysis, regional enrollment tracking, and identification of underserved areas. This supports the platform’s inclusivity objectives by revealing participation disparities across regions.

2.3.2.6 dim_loyalty_program

The loyalty program dimension captures membership program attributes:

Table 2.6: Loyalty Program Dimension Structure

Field Name	Data Type	Description
loyalty_prog_id	INT (PK)	Unique loyalty program identifier
LP_code	VARCHAR(20)	Loyalty program business code
loyalty_card	VARCHAR(20)	Membership tier (Star, Nova, Aurora)
enrollment_type	VARCHAR(30)	Method of enrollment (Online, In-person, Partner, etc.)

This dimension supports tier progression analysis, enrollment channel effectiveness evaluation, and member segmentation by loyalty level. The enrollment_type field enables assessment of which acquisition channels produce the most valuable customers.

2.3.2.7 dim_services

The services dimension categorizes ancillary services purchased by customers:

Table 2.7: Services Dimension Structure

Field Name	Data Type	Description
service_id	INT (PK)	Unique service identifier
service_name	VARCHAR(100)	Descriptive name of service
service_code	VARCHAR(20)	Business code for service

This dimension enables ancillary revenue analysis, cross-selling opportunity identification, and understanding of which additional services drive customer value. Services may include baggage handling, premium meals, seat selection, entertainment packages, and other add-ons.

2.3.3 Fact Table

The central fact table, **fact__customer__loyalty**, contains measurable events and metrics aggregating customer loyalty program activities:

Table 2.8: Fact Table Structure

Field Name	Data Type	Description
<i>Foreign Keys (Dimensions)</i>		
customer_id	VARCHAR(50) (FK)	Links to dim_customer
date_id	INT (FK)	Links to dim_dates
feedback_id	INT (FK)	Links to dim_feedback
flight_id	INT (FK)	Links to dim_flight
service_id	INT (FK)	Links to dim_services
loyalty_prog_id	INT (FK)	Links to dim_loyalty_program
<i>Measures and Metrics</i>		
total_flights	INT	Total number of flights taken
distance	DECIMAL(18,2)	Total distance traveled (kilometers)
points_accumulated	DECIMAL(18,2)	Total loyalty points earned
Points_redeemed	DECIMAL(18,2)	Points used for rewards
Dollar_Cost_Points_Redeemed	DECIMAL(18,2)	Monetary value of redeemed points
CLV	DECIMAL(18,2)	Customer Lifetime Value
Enrollment_year	INT	Year of loyalty program enrollment
Enrollment_Month	INT	Month of enrollment (1-12)
cancellation_year	INT	Year of cancellation (NULL if active)
cancellation_month	INT	Month of cancellation (NULL if active)
rating_value	DECIMAL(5,2)	Aggregated satisfaction rating
fact_code	VARCHAR(50)	Business code for fact record

Fact table grain: The fact table is designed at the *customer-activity summary* grain, meaning each row represents aggregated metrics for a customer’s interactions within a specific time period. This grain supports both detailed customer-level analysis and efficient aggregation for summary reports and dashboards.

Key measures:

- **Behavioral metrics:** total_flights and distance track travel intensity
- **Loyalty metrics:** points_accumulated and Points_redeemed measure program engagement
- **Financial metrics:** Dollar_Cost_Points_Redeemed and CLV quantify customer value
- **Temporal metrics:** Enrollment and cancellation dates enable cohort analysis
- **Satisfaction metrics:** rating_value correlates experience with behavior

Measure additivity: Most measures (points, flights, distance, dollar costs) are additive and can be summed across any dimension. CLV and rating_value are semi-additive (can be averaged but not meaningfully summed). Temporal fields support filtering and grouping but are not aggregated mathematically.

2.4 ETL Process Implementation

The Extract, Transform, and Load (ETL) process orchestrates the flow of data from CSV source files through the staging area to the data warehouse, ensuring data integrity, performing necessary transformations, and maintaining dimensional relationships.

2.4.1 ETL Architecture

The ETL process is implemented using SQL Server Integration Services (SSIS) with a modular package design:

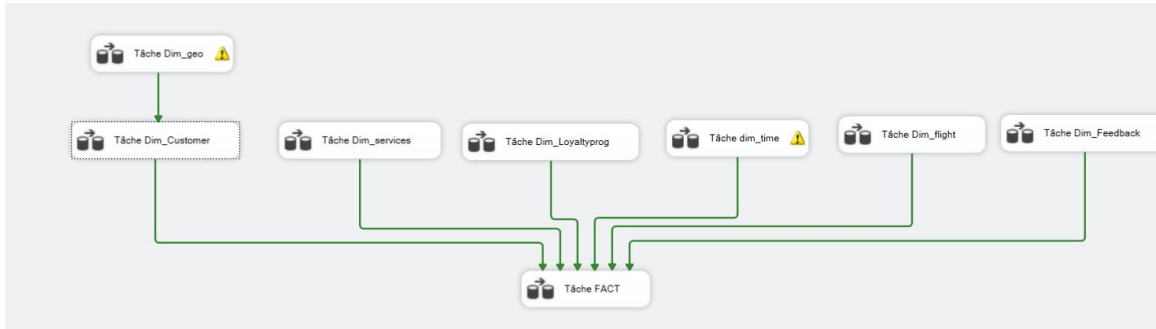


Figure 2.3: ETL Process Architecture

Package structure:

1. **Master control package:** Orchestrates the entire ETL workflow, manages execution sequence, handles error logging, and coordinates transaction boundaries
2. **CSV-to-Staging packages (4 packages):** Load each source CSV file into corresponding staging tables:
 - Load_Customer_Flight_Activity
 - Load_Customer_Loyalty_History
 - Load_Calendar
 - Load_Satisfaction_Survey
3. **Staging transformation package:** Performs data cleansing, standardization, and pre-aggregation within the staging area
4. **Dimension load packages (7 packages):** Populate each dimension table in the data warehouse
5. **Fact load package:** Populates the fact table after all dimensions are loaded, performing foreign key lookups and measure calculations

2.4.2 Stage 1: CSV to Staging Area

The first ETL stage extracts data from CSV files and loads it into the staging area:

Process steps for each CSV file:

1. **File system check:** Verify file existence in designated folder; check file accessibility and size
2. **Flat file connection configuration:**
 - Set file path and format (delimited, UTF-8 encoded)
 - Define column delimiters (comma)

- Configure text qualifiers (double quotes) for fields containing commas
- Enable header row parsing for column name extraction

3. Data flow with transformations:

- **Source:** Flat File Source component reads CSV rows
- **Data Conversion:** Convert text columns to appropriate SQL types (DT_I4 for integers, DT_R8 for decimals, DT_DBDATE for dates)
- **Derived Column:** Add load timestamp and source file name for lineage tracking
- **Conditional Split:** Route error rows to separate output
- **Destination:** OLE DB Destination loads valid rows to staging table
- **Error Output:** Redirect conversion errors to error logging table

4. Post-load validation:

- Execute SQL task to count loaded rows
- Compare count to source file line count (minus header)
- Log discrepancies for investigation

2.4.3 Stage 2: Staging Area Transformations

After CSV loading, data undergoes transformations within the staging area:

Data cleansing operations:

- **Text standardization:** UPPER() function applied to categorical fields for consistency (loyalty tiers, travel types, service classes)
- **Null handling:** Replace empty strings and placeholder values ('N/A', 'Unknown') with proper SQL NULL
- **Date standardization:** Convert various date formats to consistent SQL DATE type; handle missing dates (e.g., NULL cancellation dates for active members)
- **Numeric cleansing:** Remove currency symbols from cost fields; ensure positive values for distances and points; handle division-by-zero scenarios
- **Deduplication:** Identify duplicate customer_id entries; apply business rules to retain most recent or most complete record

Data enrichment:

- **Derived metrics:** Calculate membership duration (months between enrollment and current date or cancellation date)
- **Customer Lifetime Value (CLV) calculation:** Aggregate total spending per customer across all flights and services
- **Categorical binning:** Group continuous variables (e.g., create age groups from birth dates, salary ranges from exact salaries)
- **Geographic hierarchy validation:** Ensure city-province-country relationships are consistent; correct common misspellings

2.4.4 Stage 3: Dimension Loading

Dimensions are populated from staging tables using individual SSIS packages for each dimension:

Dimension load process:

1. **Extract distinct values:** Query staging tables for unique combinations of dimensional attributes
2. **Lookup existing records:** Check if dimension records already exist based on business keys
3. **Apply Slowly Changing Dimension (SCD) logic:**
 - **Type 1 (Overwrite):** Update existing records if attributes changed (used for most dimensions)
 - New records: Insert with new primary key
 - Unchanged records: No action required
4. **Data quality checks during load:**
 - Validate referential integrity (e.g., geo_id in dim_customer exists in dim_geography)
 - Check for required field completeness
 - Verify data type constraints
 - Ensure unique key constraints are satisfied
5. **Load confirmation:** Log row counts (inserted, updated, unchanged) and execution duration

Dimension-specific considerations:

- **dim_dates:** Pre-populated for full date range (e.g., 2010-2030); updated rarely
- **dim_customer:** Linked to dim_geography via geo_id; requires coordinated load
- **dim_geography:** Loaded first as other dimensions reference it
- **dim_loyalty_program:** Relatively static; occasional new enrollment types added
- **dim_flight, dim_services, dim_feedback:** Grow incrementally as new values appear in operational data

2.4.5 Stage 4: Fact Table Loading

The fact table is loaded last, after all dimensions are populated, ensuring all foreign keys can be resolved:

Fact load process:

1. **Aggregate staging data:** Group customer activity data by customer, computing summary metrics:
 - Sum total flights, distance, points accumulated, points redeemed
 - Calculate average satisfaction ratings

- Compute customer lifetime value
- Determine enrollment and cancellation dates

2. **Dimension key lookups:** Replace business keys with foreign keys via joins:

- Customer_id → dim_customer
- Date values → dim_dates (converted to date_pk format)
- Service attributes → dim_services
- Flight attributes → dim_flight
- Loyalty attributes → dim_loyalty_program
- Geographic identifier → dim_geography
- Feedback code → dim_feedback

3. **Handle lookup failures:**

- Log records with unmatched foreign keys
- Option 1: Create "Unknown" dimension members for missing keys
- Option 2: Exclude records from fact table and flag for investigation

4. **Fact record validation:**

- Ensure all foreign keys resolve to valid dimension records
- Validate measure ranges (e.g., Points_redeemed points_accumulated)
- Check temporal consistency (enrollment before first flight)

5. **Bulk insert:** Load validated records into fact_customer_loyalty using bulk insert for performance

6. **Post-load verification:**

- Verify referential integrity constraints
- Compare fact table row count to staging aggregation count
- Validate measure totals (sum of points, total CLV)
- Check for duplicate customer records

2.4.6 Error Handling and Logging

Comprehensive error handling ensures data quality and facilitates troubleshooting:

Table 2.9: ETL Error Handling Mechanisms

Mechanism	Description
Error output flows	SSIS data flows configured with error outputs that capture rejected rows with full column values
Error logging tables	Dedicated tables (ETL_Errors, ETL_Warnings) store error details including timestamp, package name, task name, error code, error description, and rejected data
Transaction management	Each package stage executes within a SQL transaction; automatic rollback on critical failures ensures data consistency
Checkpoint files	SSIS checkpoint files enable package restart from failure point without re-processing successful steps
Email notifications	SQL Server Agent jobs configured to send email alerts to data team on package failure or warning thresholds exceeded
Execution logs	SSIS catalog (SSISDB) maintains detailed execution history including row counts, execution duration, and performance metrics for each package run
Data quality reports	Post-ETL scripts generate quality reports with metrics on null rates, value distributions, and validation rule pass/fail counts

2.4.7 ETL Scheduling and Performance

The ETL process runs on a scheduled basis to maintain data warehouse currency:

- **Execution frequency:** Daily during off-peak hours (2:00 AM local time)
- **Execution duration:** Full refresh completes in approximately 30-45 minutes for 160,000 customer records
- **Processing strategy:** Full reload for dimensions (relatively small); incremental updates considered for fact table as data volume grows
- **Parallelization:** Independent dimension loads execute in parallel; fact load waits for all dimension completion
- **Resource management:** ETL jobs run with dedicated SQL Server agent account; resource governor limits CPU and memory usage to prevent impact on other database activities

Performance optimization techniques implemented:

- **Bulk loading:** SSIS Fast Load option enables rapid inserts using bulk insert API
- **Index management:** Non-clustered indexes dropped before fact table load, rebuilt after completion
- **Batch processing:** Large datasets processed in batches (10,000 rows per commit) to manage transaction log size
- **Minimal logging:** Database set to simple recovery model during ETL to reduce logging overhead
- **Data caching:** SSIS lookup transformations use full cache mode for small dimensions, partial cache for large dimensions

- **Filtered extractions:** Staging queries include WHERE clauses to extract only necessary data, reducing data movement

2.5 Data Quality Validation and Monitoring

After ETL completion, automated data quality checks validate the integrity, completeness, and consistency of the data warehouse:

2.5.1 Validation Framework

Table 2.10: Data Warehouse Quality Validation Checks

Check Category	Validation Rule	Action on Failure
Row count reconciliation	Fact table row count matches expected based on staging aggregations; dimension counts align with distinct source values	Alert data team; investigate discrepancies; potential reload
Referential integrity	All fact table foreign keys resolve to existing dimension records; no orphan records	Critical error; halt downstream processes; log violations
Measure completeness	Critical measures (points_accumulated, total_flights, CLV) contain no NULL values where inappropriate	Warning; flag records for review
Measure reasonableness	Measures fall within business-defined ranges (e.g., CLV > 0, points_redeemed points_accumulated, satisfaction 1-5)	Warning; create exception report for business review
Temporal consistency	Enrollment dates precede flight dates; cancellation dates follow enrollment; date keys correspond to valid calendar dates	Error log; investigate data source issues
Dimension completeness	All expected dimension members present (e.g., all loyalty tiers Star/Nova/Aurora exist)	Alert if core dimension values missing
Duplicate detection	No duplicate records in dimensions based on business keys; no duplicate customer entries in fact table	Critical error; deduplication required
Distribution checks	Key metric distributions match expected patterns (e.g., loyalty tier distribution roughly 50%/30%/20%)	Warning if significant deviation from baseline

2.6 Data Lineage and Metadata Management

To support maintainability, auditability, and regulatory compliance, comprehensive metadata and lineage documentation is maintained:

2.6.1 Technical Metadata

- **Source-to-target mappings:** Documentation of how each data warehouse field derives from source CSV files, including:
 - Source file name and column name
 - Target table and column name
 - Transformation logic applied (if any)
 - Business rules governing mapping
- **Transformation catalog:** Repository of all data transformations including:

- Calculation formulas (e.g., CLV computation)
- Aggregation logic (e.g., total flights per customer)
- Derived column logic (e.g., membership duration)
- Cleansing rules (e.g., standardization of categorical values)
- **Data lineage diagrams:** Visual representation of data flow from source CSV files through staging to final warehouse tables, showing all intermediate transformations
- **Schema documentation:** Complete data dictionary including:
 - Table descriptions and business purpose
 - Column definitions and data types
 - Primary and foreign key relationships
 - Index definitions and rationale
 - Constraints and validation rules

2.6.2 Business Metadata

- **Business glossary:** Definitions of business terms in plain language (e.g., "What is Customer Lifetime Value?" "How is churn defined?")
- **Measure definitions:** Detailed explanations of how each metric is calculated and what it represents
- **Dimension hierarchies:** Documentation of logical hierarchies (e.g., City → Province → Country in geography)
- **Business rules:** Catalog of business logic embedded in ETL processes (e.g., "Active customer has no cancellation date")

2.6.3 Operational Metadata

- **ETL execution history:** Historical log of all ETL runs including start time, end time, duration, rows processed, errors encountered
- **Data refresh schedule:** Documentation of when each table was last refreshed and next scheduled refresh
- **Change log:** Version history of ETL packages, data warehouse schema changes, and business rule modifications
- **Access audit trail:** Record of who accessed data warehouse, when, and what queries were executed (for security and compliance)

This metadata supports impact analysis when source data structures change, facilitates onboarding of new team members, and provides audit trail for regulatory requirements.

Conclusion

This chapter presented the data integration and preparation foundation of the Airlytics platform. Starting from four source CSV files containing information on 160,000 customers, the chapter detailed the complete data pipeline from raw files through staging area consolidation to a dimensional data warehouse optimized for analytics.

The star schema design with seven dimension tables (customer, dates, feedback, flight, geography, loyalty program, services) and one central fact table (fact_customer_loyalty) provides a flexible, high-performance foundation for the analytical models and business intelligence dashboards described in subsequent chapters. The ETL processes ensure daily data refresh while maintaining data quality through comprehensive validation and error handling mechanisms.

Key accomplishments of this phase include:

- Successful integration of 160,000 customer records from disparate CSV sources
- Implementation of robust staging area for data consolidation and cleansing
- Design and deployment of dimensional data warehouse following star schema best practices
- Development of automated ETL pipelines with comprehensive error handling
- Establishment of data quality framework with ongoing monitoring
- Documentation of complete data lineage and metadata for maintainability

Together, these data preparation components transform raw operational CSV files into a structured analytical asset ready to support the strategic objectives outlined in Chapter 1. The next chapter will present the machine learning models and advanced analytics that leverage this data foundation to generate predictive insights for customer churn, redemption behavior, lifetime value estimation, and customer segmentation.

Chapter 3

Data Preparation and Machine Learning Pipeline: Model Development and Validation

3.1 Introduction

This report presents the complete machine learning pipeline implemented in the Airlytics platform, encompassing both data preparation and predictive model development. The pipeline transforms the structured data warehouse into specialized datasets tailored for specific analytical objectives, then applies advanced machine learning techniques to generate actionable predictions.

The report is organized into two major parts:

- **Part I:** Data preparation pipeline that extracts, cleanses, enriches, and consolidates warehouse data
- **Part II:** Development, training, and validation of six specialized machine learning models

3.2 PART I: Data Preparation Pipeline

3.2.1 Pipeline Overview and Objectives

The data preparation pipeline creates a reproducible, traceable process that transforms raw data warehouse tables into six specialized datasets aligned with strategic objectives:

1. Churn Prediction (Binary Classification + Time-to-Churn Regression)
2. Tier Progression Prediction (Probability Scoring)
3. Loyalty Points Utilization (Clustering + Classification + Forecasting)
4. Customer Lifetime Value (CLV) Calculation and Prediction
5. Traveler Profiling, Clustering and Seasonal Prediction
6. Geographic Equity Analysis

3.2.2 Data Extraction and Versioning

3.2.2.1 Data Warehouse Connection

The pipeline establishes a secure connection to the Airline Data Warehouse using Windows Authentication via ODBC Driver 17 for SQL Server. The connection leverages built-in Windows credentials, eliminating the need for separate credential management.

Connection Parameters:

- **Driver:** ODBC Driver 17 for SQL Server
- **Server:** ADHAM (local domain server)
- **Database:** AIRLINE_DW
- **Authentication:** Windows (Trusted Connection)
- **Technology:** Python pyodbc library for query execution

3.2.2.2 Data Extraction and Initial Archival

Eight core tables were extracted from the data warehouse, comprising 7 dimension tables and 1 fact table:

Extracted Tables:

- **fact_customer_loyalty:** Primary fact table containing loyalty transactions and behavioral records
- **dim_customer:** Customer demographic and enrollment information (Customer ID, enrollment dates, cancellation tracking)
- **dim_feedback:** Customer satisfaction ratings and feedback data
- **dim_flight:** Flight transaction details (flight frequency, distance, class of service)
- **dim_geography:** Geographic information (country, province, city, postal code)
- **dim_loyalty_program:** Loyalty program tier, points accumulated, and points redeemed
- **dim_services:** Service utilization and ancillary service engagement
- **dim_dates:** Temporal dimension supporting temporal analysis and aggregation

Archival Strategy: Data was versioned through dual archival:

- **Excel Multi-Sheet Archive** (DW_airline_dump.xlsx): Each dimension table saved as a separate sheet for reference and manual inspection
- **CSV Export:** Fact table exported as CSV for rapid reload without database connectivity

This dual approach ensures reproducibility while maintaining audit trail and reducing dependency on live database availability.

3.2.3 Data Integration and Consolidated Master Table

3.2.3.1 Star Join Architecture

The seven dimension tables were sequentially merged with the fact table using a star schema join pattern, creating a denormalized master table suitable for machine learning:

Join Sequence:

1. Base: fact_customer_loyalty (starting point)
2. Merge 1: + dim_customer (customer demographics)
3. Merge 2: + dim_flight (flight behavior)
4. Merge 3: + dim_geography (location data)
5. Merge 4: + dim_loyalty_program (tier and points)
6. Merge 5: + dim_feedback (satisfaction metrics)
7. Merge 6: + dim_services (service engagement)
8. Merge 7: + dim_dates (temporal features)

Result: Consolidated master table containing $n = 8,742$ customer records with 28 features describing:

- Customer demographics (gender, marital status, education, salary)
- Loyalty program status (tier, points accumulated, points redeemed)
- Travel behavior (total flights, flight distance, flight month, class)
- Enrollment lifecycle (enrollment dates, cancellation status, duration)
- Satisfaction and engagement (ratings, feedback, service usage)
- Geographic information (country, province, city, postal code)

3.2.4 Data Quality Assessment and Cleaning

3.2.4.1 Preliminary Analysis

Initial data exploration revealed key characteristics:

3.2.4.2 Handling Missing Values

Missing value handling was deferred to objective-specific data preparation:

- **For Churn Modeling:** Cancellation months for non-churned customers imputed with reference date of January 2019
- **For Tier Progression:** Rows with missing values in predictor variables dropped (limited impact given sample size)
- **For Geographic Analysis:** Missing postal codes retained; geographic location preserved at country/province level

Table 3.1: Data Quality Assessment Summary

Characteristic	Finding	Treatment Decision
Skewed Distributions	CLV, points, distance highly right-skewed	Retained; indicates legitimate extreme customers
Business Zeros	Zero flights, zero points, inactive accounts	Retained; represent genuine customer states
Outliers	High-value customers (CLV > \$100k)	Retained; valuable customer segment
Missing Values	Cancellation dates (inactive), optional demographics	Handled per objective (imputation vs. removal)
Duplicates	No duplicate customer records detected	None to remove

3.2.5 Categorical Variable Encoding Strategy

Machine learning models require numerical representations of categorical variables. Three encoding strategies were applied based on variable characteristics and downstream model requirements:

Table 3.2: Categorical Encoding Strategy by Variable

Encoding Type	Variables Applied	Classes	Rationale
Ordinal	education_encoded, class_encoded, tier_encoded	3-6 levels	Preserves natural hierarchy (e.g., education level, cabin class)
Label	gender_encoded, marital_status_encoded, enrollment_type_encoded	2-3 categories	Binary or few-class variables; suitable for tree-based models
Binary	travel_type_encoded	2 categories	Binary variable (domestic vs. international)
Target Encoding	postal_code_encoded	Postal regions	Geographic granularity while reducing dimensionality

Encoding Implementation:

- **Scikit-learn LabelEncoder:** Used for label and ordinal encoding
- **Custom Ordinal Mapping:** Manual mapping for inherently ordered variables
- **Preservation Strategy:** Original categorical columns retained in archive for interpretation; only encoded versions used in modeling

3.2.6 Objective-Specific Feature Engineering and Dataset Creation

Following the consolidated master table creation and cleaning, six objective-specific datasets were generated through targeted feature engineering:

3.2.6.1 Dataset 1: Churn Prediction (DS1_Churn)

Target Variables:

- **churn** (binary): Derived from `cancellation_year` field
 - $churn = 1$ if customer cancelled ($cancellation_year \neq -1$)
 - $churn = 0$ if customer remains active ($cancellation_year = -1$)
- **enrollment_duration** (continuous): Months between enrollment and cancellation
 - For churned customers: actual months elapsed
 - For non-churned customers: imputed reference date of January 2019

Data Leakage Prevention: Temporal columns (`enrollment_month`, `enrollment_year`, `cancellation_month`, `cancellation_year`) are removed after target creation to prevent models from using direct churn information.

Feature Set Size: 16 features (classification), 21 features (regression)

Feature Selection Approach: Correlation-based filtering with thresholds:

- Minimum correlation with target: $|r| \geq 0.05$ (retain features with meaningful signal)
- Maximum inter-feature correlation: $r_{pairwise} \leq 0.95$ (reduce multicollinearity)

Final Dataset Size: $n = 8,742$ samples

3.2.6.2 Dataset 2: Tier Progression (DS2_Tier)

Target Variable: `upgrade_potential` (binary classification)

- $upgrade = 1$ if customer moved to higher tier
- $upgrade = 0$ if customer remained at current tier

Key Features:

- Current tier, points accumulated, flight frequency, cabin class
- Customer satisfaction (rating), salary, CLV
- Feature engineering: `avg_points_per_flight`, `redemption_ratio`, `clv_per_flight`, `long_distance_flag`

Sample Characteristics: Balanced classes through stratified sampling

3.2.6.3 Dataset 3: Loyalty Points Utilization (DS3_Loyalty)

Target Variables:

- **redeemed_flag** (binary): 1 if customer redeemed points, 0 otherwise
- **redemption_rate** (continuous): Points redeemed / Points accumulated
- **high_redeemer** (binary): Customers above 50th percentile of redemption rate

Key Features:

- Points accumulated and redeemed

- Tier level and loyalty program tenure
- Travel frequency and distance
- Enrollment and cancellation timing
- Satisfaction metrics

3.2.6.4 Dataset 4: Customer Lifetime Value (DS4_CLV)

Target Variable: CLV (continuous) and clv_category (ordinal)

- CLV directly available in master table
- Category mapping: Low (\$0–\$20k), Medium (\$20k–\$50k), High (>\$50k)

Key Features:

- Historical CLV (lagged), flight frequency, distance
- Tier progression history, tenure, satisfaction
- Geographic and demographic factors

3.2.6.5 Dataset 5: Traveler Profiling and Clustering (DS5_Clustering)

Clustering Features: Unsupervised dataset with no target variable

- Total_Flights: travel intensity
- Distance: long-haul vs. short-haul patterns
- Flight_Month: seasonality
- Points_Redeemed, Dollar_Cost_Points_Redeemed: loyalty engagement
- Cancellation_Month, Cancellation_Year: temporal patterns
- Enrollment_Encoded: program tenure

Dataset Size: Full master table ($n = 8,742$)

3.2.6.6 Dataset 6: Geographic Equity (DS6_Geographic)

Aggregation Levels: Country, Province, City

Key Metrics Computed:

- Enrollment rate by geography
- Average CLV and engagement metrics by region
- Churn rate disparities across geographic areas
- Service availability and tier distribution

3.2.7 Final Exported Datasets

All objective-specific datasets exported as CSV files for direct loading in modeling notebooks:

- cleaned_master2.csv (consolidated master table, 8,742 rows, 28 features)
- churn_training.csv / survival_training.csv (DS1)
- tier_progression_training.csv (DS2)
- data_redeemed.csv (DS3 for classification)
- data_reg.csv (DS3 for time series)

Objectives 4–6

3.3 PART II: Machine Learning Models Implementation

This section presents the development, training, and validation of six machine learning models corresponding to strategic objectives. Each subsection details the model approach, key techniques, performance evaluation, and business insights.

3.3.1 Objective 1: Customer Churn Prediction

3.3.1.1 Overview

Customer churn prediction addresses two complementary machine learning tasks structured in three phases:

1. **Phase 1:** Feature selection by objective
2. **Phase 2:** Binary classification of churn status
3. **Phase 3:** Regression-based time-to-churn prediction

3.3.1.2 Phase 1: Feature Selection by Objective

Starting from a cleaned dataset with $n = 8,742$ samples across $p = 28$ features, exploratory analysis focused on:

- Correlation analysis between candidate features and target variables
- Redundancy elimination to reduce multicollinearity
- Business alignment ensuring operational relevance

Target Variable Engineering:

Churn binary target created from cancellation year field, with enrollment duration calculated as temporal difference between cancellation and enrollment months. For non-churned customers, a reference cancellation date of January 2019 was imputed.

Classification Feature Set: 16 predictors retained with absolute correlation threshold of 0.05, designed to capture churn propensity at subscription time.

Regression Feature Set: 21 predictors retained with lower correlation threshold to capture weak signals, prioritizing features related to customer lifecycle activity.

3.3.1.3 Phase 2: Binary Classification – Churn vs. Non-Churn

Objective: Classify customers into binary categories (churned vs. active) for early identification of at-risk customers.

Methodology:

- Data balancing: SMOTE (0.5 strategy) with RandomUnderSampler (1.0 strategy)
- Feature scaling: StandardScaler normalization to zero mean and unit variance
- Algorithm evaluation: Three algorithms trained on identical balanced dataset
- Hyperparameter optimization: Cross-validation with F1-weighted scoring
- Train-test split: 80-20 stratified split preserving class distribution

Algorithms Evaluated:

- Distance-Based: K-Nearest Neighbors with optimal k selection via cross-validation
- Ensemble Methods: Random Forest with hyperparameter grid search
- Gradient Boosting: XGBoost with grid search optimization

Performance Metrics: Accuracy, Balanced Accuracy, Precision, Recall, F1-Score, and ROC-AUC used to evaluate both class-level and overall performance.

Table 3.3: Classification Model Performance Comparison

Algorithm	Accuracy	Precision	Recall	F1-Score	ROC-AUC
K-Nearest Neighbors ($k = 3$)	0.8425	0.8437	0.8425	0.8424	0.9051
Random Forest	0.7810	0.7897	0.7810	0.7794	0.8811
XGBoost	0.7159	0.7462	0.7159	0.7069	0.8216

Final Model Selection: K-Nearest Neighbors ($k = 3$) emerged as the superior classification model, achieving the highest performance across all evaluation metrics.

KNN Model Performance Summary:

Table 3.4: Optimized K-Nearest Neighbors Classification Results

Metric	Value
Accuracy	0.8425
Balanced Accuracy	0.8425
Precision	0.8437
Recall	0.8425
F1-Score	0.8424
ROC-AUC	0.9051

Rationale: KNN achieves superior churn detection across all metrics, with particularly strong ROC-AUC (0.9051) indicating excellent class separability. The model demonstrates balanced sensitivity between churn (recall: 0.87) and non-churn (recall: 0.81) customers, minimizing both false positives and false negatives in high-risk customer segments.

3.3.1.4 Phase 3: Regression – Time-to-Churn Prediction

Objective: Predict continuous variable *enrollment duration until customer churn* (measured in months) to enable temporal risk stratification:

- Immediate Risk: ≤ 6 months
- Short-Term Risk: 6-12 months
- Medium-Term Risk: 12-24 months
- Long-Term Stability: > 24 months

Methodology:

- Data preprocessing: Loading from *survival_training.csv* with missing value removal
- Feature selection: SelectKBest applied, retaining approximately 60% of original features
- Normalization: StandardScaler with fit on training set only (preventing data leakage)
- Train-test split: 80-20 random split
- Hyperparameter optimization: Grid search with cross-validation for each algorithm

Algorithms Evaluated:

- Ensemble Methods: Random Forest, Gradient Boosting, XGBoost
- Optimized Gradient Boosting: LightGBM with early stopping

Performance Metrics: R^2 Score (variance explained), Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE).

Table 3.5: Regression Model Baseline Performance Comparison

Algorithm	R^2 Score	MAE (months)	RMSE (months)	MAPE (%)
Random Forest	0.7156	6.34	10.87	19.2
XGBoost	0.7428	5.89	10.15	17.8
Gradient Boosting	0.7512	5.67	9.92	17.1

Final Model Selection: Gradient Boosting emerged as the baseline leader. Further optimization via feature selection and LightGBM implementation yielded significant improvements.

Table 3.6: Optimized LightGBM Regression Model Results

Dataset	R^2 Score	MAE (months)	RMSE (months)	MAPE (%)
Training Set	0.8542	4.87	7.92	14.6
Test Set	0.8321	5.20	8.54	15.3
Improvement vs. Baseline	+11.0%	-8.3%	-13.9%	-10.5%

Model Selection Rationale: LightGBM Regressor designated as production model based on:

- Predictive accuracy: 83.21% variance explained on test set (+11.0% vs. Gradient Boosting baseline)
- Operational reliability: 5.20-month average prediction error (-8.3% reduction)
- Minimal overfitting: $\Delta R^2 = 0.0221 < 0.05$ (well-calibrated model)
- Feature efficiency: 40% reduction in input dimensionality via SelectKBest
- Computational efficiency: Execution time $\approx$ 60-90 seconds for full training pipeline (10-15x faster than traditional Gradient Boosting)

3.3.1.5 Integrated Churn Reduction Strategy

The complementary classification and regression models form an integrated risk assessment system:

1. **Stage 1:** K-Nearest Neighbors classifier identifies at-risk customers (threshold: churn probability > 0.5)
2. **Stage 2:** LightGBM regressor predicts time horizon for classified churners
3. **Stage 3:** Retention campaigns prioritized by temporal risk tier and prediction confidence

3.3.1.6 Conclusion

The three-phase machine learning initiative establishes a data-driven churn reduction framework combining objective-driven data balancing and feature scaling, binary classification achieving 84.25% accuracy with 90.51% ROC-AUC, and regression-based time-to-churn prediction explaining 83.21% of enrollment duration variance with minimal overfitting and significant computational efficiency gains.

3.3.2 Objective 2: Tier Progression Prediction and Accelerator Identification

3.3.2.1 Overview

Tier progression prediction encompasses two complementary analytical approaches:

1. **Regression Analysis:** Predict probability of tier progression using customer behavioral and demographic features
2. **Classification Analysis:** Identify key accelerators driving tier upgrades

3.3.2.2 Regression Analysis – Tier Progression Probability

Objective: Evaluate multiple regression models to predict the probability of tier progression using features such as satisfaction, CLV, rating, salary, education, marital status, class, cancellation timing, travel type, and geographic location.

Models Tested: Logistic Regression, Random Forest, and Gradient Boosting.

Key Findings:

Logistic Regression with GridSearchCV:

- Optimized with penalties (L2, ElasticNet) and C values from 0.1 to 10
- AUC OvR: 0.9072, Progression Score: 0.7701
- PowerTransformer + PolynomialFeatures (degree 2) + StandardScaler pipeline captured non-linear relationships
- Performance robust but slightly lower than ensemble methods

Random Forest:

- Ensemble of 500 trees with max_depth=14, min_samples_leaf=2, and balanced class weights
- AUC OvR: 0.9628 (highest), Progression Score: 0.8215
- Superior ability to identify progression-ready customers through parallel tree construction
- Excellent balance across all three tiers

Gradient Boosting:

- Sequential boosting with 300 estimators, learning_rate=0.1, and shallow trees (max_depth=3)
- AUC OvR: 0.9442, Progression Score: 0.7710
- Iterative error correction provided strong predictions comparable to Logistic Regression

Best Model: Random Forest with AUC OvR of 0.9628 and Progression Score of 0.8215, demonstrating highest discrimination capability.

3.3.2.3 Classification Analysis – Tier Upgrade Accelerators

Objective: Apply classification algorithms to identify key accelerators of tier progression using behavioral features such as class preference, CLV, satisfaction, salary, points accumulation patterns, and flight distance.

Models Tested: Random Forest with GridSearchCV optimization and Gradient Boosting.

Key Findings:

Random Forest with GridSearchCV:

- Optimized ensemble of 450 trees with max_depth=16, min_samples_leaf=2, and balanced class weights
- Test AUC: 0.9601 (highest), Accuracy: 0.84
- GridSearchCV tested 18 parameter combinations across 5 folds (90 fits total)
- Feature engineering included avg_points_per_flight, redemption_ratio, clv_per_flight, and long_distance_flag
- Permutation importance revealed class_encoded (0.289) and CLV (0.122) as dominant accelerators

Gradient Boosting:

- Sequential boosting with 300 estimators, learning_rate=0.1, and shallow trees (max_depth=3)
- Test AUC: 0.9406, Accuracy: 0.81
- Permutation importance confirmed class_encoded (0.280) and CLV (0.124) as top predictors
- Performance slightly below Random Forest but with faster training time

Best Model: Random Forest with Test AUC of 0.9601 and accuracy of 0.84.

Top 5 Accelerators: class_encoded, CLV, clv_per_flight, satisfaction, salary.

3.3.2.4 Conclusion

Random Forest delivered superior performance (AUC 0.9601) compared to Gradient Boosting (AUC 0.9406). Permutation importance analysis identified class preference and customer lifetime value as the dominant accelerators of tier progression, explaining over 41% of predictive power combined. All models demonstrated strong discriminatory power (AUC > 0.90), confirming that the selected features are highly predictive of tier advancement. Engineered features enhanced model interpretability and confirmed that accumulation efficiency matters more than raw volume for tier advancement.

3.3.3 Objective 3: Loyalty Points Utilization

3.3.3.1 Overview

Loyalty Points Utilization encompasses three complementary machine learning objectives to maximize customer engagement with the loyalty program and optimize point inventory management:

1. **Phase 1:** Cluster customers based on their redemption behaviors to identify groupings
2. **Phase 2:** Classify customers based on their likelihood to redeem loyalty points
3. **Phase 3:** Forecast loyalty point utilization trends to identify seasonality

This integrated approach enables segment-specific engagement strategies, personalized targeting, and data-driven loyalty program management.

3.3.3.2 Phase 1: Customer Clustering by Redemption Behavior

Objective: Identify distinct customer segments based on loyalty program engagement and redemption patterns through unsupervised learning.

Approach:

- **Task type:** Unsupervised learning (customer clustering)
- **Goal:** Identify three or more meaningful and interpretable customer clusters
- **Target:** Understand customer loyalty patterns and segment-specific redemption behaviors

Clustering Algorithms Evaluated:

- **K-Means:** Scalable and interpretable for large datasets, suitable for business dashboards
- **Hierarchical Clustering (CAH):** Used for exploratory analysis and validation of cluster structure

K-Means was retained as the final clustering method due to its stability, interpretability, and suitability for integration into business intelligence dashboards and applications.

Feature Selection: Behavioral, temporal, and loyalty-related variables:

- **Total_Flights:** Overall travel frequency
- **Distance:** Travel distance patterns (short-haul vs. long-haul)
- **Flight_Month:** Seasonality and periodic travel behavior
- **Points_Redeemed:** Loyalty program engagement intensity
- **Dollar_Cost_Points_Redeemed:** Monetary value of redeemed rewards
- **Cancellation_Month:** Cancellation timing and behavioral stability
- **Cancellation_Year:** Long-term cancellation behavior
- **Enrollment_Encoded:** Loyalty program tenure and maturity

Optimal Cluster Determination:

- Elbow method to analyze inertia variation across different cluster numbers
- Silhouette score to assess cluster cohesion and separation
- Business interpretability assessment to ensure meaningful customer segments

Identified Customer Segments:

- **Active but Low-Intensity Flyers**

- Fly occasionally
- Accumulate points slowly
- Rarely redeem points

High redemption thresholds in airline loyalty programs discourage frequent redemption.

- **Redemption-Focused Members**

- Low flight activity (few flights / short distances)
- High number of points redeemed

This behavior is often driven by *partner-only earning* programs.

Example: Co-branded airline–bank credit cards:

- Customers open a bank account and receive a co-branded credit card
- Card spending earns loyalty points (e.g., \$1 = 1 point)
- Points are redeemed quickly, without frequent flying

Validation: Profiles were validated using quantitative clustering metrics (silhouette score) and qualitative interpretability analysis, ensuring each cluster represents a coherent and meaningful customer segment.

Applications:

- Customer profiling and redemption behavior analysis
- Segment-specific engagement strategies tailored to redemption propensity
- Power BI dashboards with segment-based metrics and KPIs
- Foundation for targeted activation campaigns and retention interventions

3.3.3.3 Phase 2: Classification – Likelihood to Redeem Loyalty Points

Objective: Build predictive models to classify customers based on their likelihood to redeem loyalty points, enabling proactive engagement strategies and personalized offers.

Data Preparation:

- Target variable: **redeemed_flag** (1 = redeemed points, 0 = did not redeem)
- Class imbalance handling: RandomOverSampler applied to balance minority class (redeemers)
- Imbalance ratio: Original dataset heavily skewed toward non-redeemers

Feature Set:

- **Points__Accumulated:** Total loyalty points earned by customer
- **Enrollment__Month/Year:** Customer enrollment timing and program tenure
- **Cancellation__Month:** Churn timing indicator
- **Total__Flights:** Number of flights taken (engagement proxy)
- **Distance:** Total distance traveled (customer value indicator)

Preprocessing Pipeline:

- RandomOverSampler: Balance classes by replicating minority samples
- Train-test split: 80-20 stratified split after oversampling
- MinMaxScaler: Normalize features to [0, 1] range for improved model convergence
- Stratification: Maintain class distribution in both train and test sets

Models Evaluated:

- K-Nearest Neighbors (k=3): Simple distance-based baseline
- Linear SVM with balanced class weights: Support vector classification
- Decision Tree (max_depth=10): Tree-based single learner
- Random Forest (n_estimators=300, balanced weights): Ensemble tree method
- XGBoost (n_estimators=300, max_depth=6): Gradient boosted ensemble

Table 3.7: Classification Model Performance – Redemption Likelihood

Model	Accuracy	Precision	Recall	F1-Score	AUC
KNN (k=3)	0.9628	0.9379	0.9913	0.9639	0.9845
Linear SVM	0.5388	0.5369	0.5645	0.5504	N/A
Decision Tree	0.5583	0.6535	0.2485	0.3600	0.5844
Random Forest	0.9827	0.9667	0.9998	0.9830	0.9961
XGBoost	0.6194	0.6874	0.4381	0.5351	0.6886

Best Model: Random Forest with 98.27% accuracy, 99.98% recall, and 0.9961 AUC, ensuring comprehensive identification of redemption-ready customers with minimal false negatives.

3.3.3.4 Phase 3: Forecasting – Seasonality in Loyalty Point Utilization

Objective: Forecast loyalty point utilization trends to identify seasonality patterns and support inventory planning for point-based rewards and capacity management.

Data Preparation:

- Time series: Daily points redemption aggregated to monthly frequency
- Data frequency: Monthly aggregation for trend and seasonality analysis
- Stationarity testing: Assessed for appropriate modeling approach
- Missing value treatment: Minimal missing data in aggregated time series

Time Series Characteristics:

- Dataset size: Multiple years of monthly redemption history
- Trend component: Gradually increasing or stable redemption volumes
- Seasonal component: Clear 12-month seasonal patterns
- Volatility: Moderate variability with identifiable peak periods

Forecasting Models Evaluated:

1. SARIMA (Seasonal AutoRegressive Integrated Moving Average):

- **Parameters:** order=(1,1,1), seasonal_order=(1,1,1,12)
- **Approach:** Explicitly models seasonal patterns with 12-month seasonal cycle
- **Training:** Fit on historical data with 80-20 train-test split
- **Strengths:**
 - Interpretable ARIMA parameters (AR, differencing, MA order)
 - Efficient with limited historical data
 - Captures autocorrelation and moving average terms explicitly
- **Output:** Point forecasts with confidence intervals

2. Prophet (Facebook’s Time Series Forecasting):

- **Configuration:** Yearly seasonality enabled, weekly/daily disabled
- **Approach:** Flexible decomposition of trend, seasonality, and holidays
- **Training:** Automatic trend changepoint detection and seasonal fitting
- **Strengths:**
 - Robust to missing data and outliers
 - Flexible seasonal decomposition with multiple components
 - Intuitive interpretation of trend and seasonal patterns

- Built-in capability for holiday effects and special events

- **Output:** Probabilistic forecasts with 95% confidence intervals

Model Training: Both models trained on historical monthly redemption data with identical 80-20 train-test split for fair comparison.

Table 3.8: Time Series Forecasting Model Comparison

Model	MAE	RMSE	R ² Score	Relative Error (%)
SARIMA	8,971.76	9,911.27	0.9565	4.96%
Prophet	9,826.65	10,703.33	0.9493	5.43%

Best Model: SARIMA with R² of 0.9565, MAE of 8,971.76, and relative error of 4.96%, providing accurate seasonal forecasts with interpretable parameters for operational planning.

3.3.3.5 Integrated Loyalty Points Utilization Strategy

The three-phase approach creates a comprehensive loyalty program optimization framework:

1. **Stage 1:** K-Means clustering identifies distinct redemption-based customer segments
2. **Stage 2:** Random Forest classifier predicts individual redemption likelihood for targeted engagement
3. **Stage 3:** SARIMA forecasting enables seasonal campaign and inventory planning

Integrated Workflow:

- **Customer Segmentation:** Use clustering results to create segment-specific engagement strategies
- **Targeting:** Apply classification model to identify high-redemption-likelihood customers within each segment
- **Planning:** Use seasonal forecasts to time campaigns and promotions optimally
- **Optimization:** Adjust offer strategies based on segment-level redemption patterns and seasonal trends

Expected Business Impact:

- Improved program engagement through segment-specific activation
- Optimized point liability management through accurate demand forecasting
- Enhanced customer satisfaction through timely, relevant point-based offers
- Operational efficiency through demand-driven inventory and resource planning
- Data-driven loyalty program management enabling continuous optimization

3.3.3.6 Conclusion

The three-phase Loyalty Points Utilization framework provides a unified and data-driven approach to improving customer engagement and optimizing loyalty program management. K-Means clustering identified distinct customer segments based on redemption behavior, enabling meaningful segmentation of loyalty program users. Random Forest classifier achieved 98.27% accuracy for predicting customers' likelihood to redeem loyalty points, supporting targeted and personalized engagement strategies. SARIMA forecasting effectively captured seasonal patterns in loyalty point utilization with 95.65% variance explained, allowing accurate anticipation of redemption demand. The integration of segmentation, classification, and forecasting enables data-driven decision-making, improved campaign timing, and optimized loyalty program performance.

3.3.4 Objective 4: Customer Lifetime Value (CLV) Prediction

3.4.1 Predictive Modeling

CLV prediction forecasts total customer revenue to enable value-based segmentation and strategic resource allocation through regression analysis.

3.4.2 Feature Selection & Exploratory Analysis

Dataset: n = 44,663 samples, 8 features

Selected Features:

- Rating_value, Salary, Education_encoded, Marital_status_encoded, Class_encoded, Tier_encoded, Satisfaction

Data Preprocessing:

- StandardScaler normalization
- Train-test split: 70-30 stratification

3.4.3 Algorithm Evaluation

Algorithms Tested: Ridge Regression, XGBoost, Random Forest

Table 1: CLV Model Performance

Algorithm	RMSE (test)	R ² (test)	MAE
Ridge Regression	6,664.95	0.0496	5,847.23
XGBoost	3,408.27	0.7432	2,156.44
Random Forest	2,844.52	0.8196	1,923.87

3.4.4 Selected Model & Results

Model: Random Forest (2,000 trees, unlimited depth, min_samples_split=2)

Performance:

- R² Score: 0.8196 (82% variance explained)
- RMSE: \$2,844.52
- MAE: \$1,923.87

3.4.5 Business Strategy

Value Tiers:

- **Top 20% (CLV > \$8,500):** VIP retention, premium upgrades
- **Middle 50% (CLV \$4,000–\$8,500):** Tier acceleration, service improvements
- **Bottom 30% (CLV < \$4,000):** Entry pricing, engagement bundling

Expected Impact: 60% budget to top 20% improves retention ROI by 2.5–3.2x; cross-sell increases premium conversion by 25–35%

3.4.6 Conclusion

Random Forest achieves 82% CLV prediction accuracy, enabling value-based resource allocation and targeted retention campaigns with <50ms per prediction.

3.3.4.1 3.5.1 Clustering

Clustering identifies natural customer groupings for targeted marketing and personalized service delivery.

3.5.2 Algorithm Evaluation

Algorithms Tested: K-Means (k=2–5), DBSCAN, Agglomerative Clustering, GMM

Table 2: Clustering Performance (Silhouette Score)

Algorithm	Config	Silhouette	Interpretability
K-Means	k=2	0.198	High
K-Means	k=3–5	0.190–0.210	Poor
DBSCAN	eps=1.6	0.200	Poor
Agglomerative	k=4, avg	0.185	Poor
GMM	k=2	0.191	Moderate

Selection Rationale: K-Means k=2 selected for superior business interpretability despite modest silhouette score; clear visual coherence and strong feature differentiation.

3.5.3 Selected Model & Cluster Profiles

Model: K-Means (k=2, random_state=42)

Cluster Composition:

Segment	Count	%	Mean CLV	% Revenue
Cluster 0: Premium Loyal	19,396	43.4%	\$7,244	55%
Cluster 1: Basic Economy	25,267	56.6%	\$5,892	45%

Table 3: Cluster Characteristics

Feature	Cluster 0	Cluster 1	Gap
Rating	3.8/5.0	2.9/5.0	+31%
Premium Class %	68%	22%	+46pp
High Tier %	72%	28%	+44pp
Mean Salary	\$65,200	\$64,800	0.6%

Cluster 0 Profile: Higher satisfaction (+31%), premium class preference (68%), high loyalty (72% Aurora/Nova), superior CLV (+23%)

Cluster 1 Profile: Lower satisfaction (24%), economy focus (78%), basic tier (85% Star), growth opportunity

3.5.4 Segmentation Strategy

Budget Allocation:

- **Cluster 0 (43%, 55% revenue):** VIP programs, premium service guarantees (55% retention budget)
- **Cluster 1 (57%, 45% revenue):** Tier acceleration, service improvements (45% growth budget)

Expected Impact: Addressing Cluster 1 satisfaction gaps (24%) through service improvements could reduce churn by 10–15%, yielding 2–3% CLV growth; tier acceleration programs could migrate 15–20% to higher tiers (+\$1,000–\$2,000 per customer)

3.5.5 Conclusion

K-Means segmentation divides 44,663 customers into two actionable segments aligned with CLV predictions. Clear feature differentiation on satisfaction, class preference, and tier status enables targeted campaigns improving retention ROI by 2.5–3.2x. Both models ready for immediate CRM integration with quarterly retraining recommended.

3.3.5 Objective 5: Enrollment Insights

3.3.5.1 Overview

Enrollment Insights encompasses two complementary machine learning objectives designed to increase loyalty program participation and optimize member engagement through data-driven customer understanding:

1. **Phase 1:** Segment customers based on region, income, and education to identify high-potential groups for targeted acquisition
2. **Phase 2:** Classify customers based on their likelihood to become highly engaged members

This integrated approach enables targeted marketing strategies, personalized acquisition campaigns, and proactive identification of high-value prospects to drive a 10% increase in registration rates over three years.

3.3.5.2 Phase 1: Customer Segmentation Analysis

Objective: Segment customers based on geographic and socio-economic characteristics to identify high-potential groups for targeted marketing and equitable program expansion.

Approach:

- **Task type:** Unsupervised learning (customer segmentation)
- **Goal:** Identify three to five meaningful and interpretable customer segments
- **Target:** Understand regional and socio-economic patterns driving enrollment and participation

Data Exploration and Understanding:

Initial exploratory data analysis revealed key distribution patterns:

- **Province Distribution:** Ontario dominates with 5,404 customers (32.3%), followed by British Columbia (4,409, 26.3%) and Quebec (3,300, 19.7%); underrepresented regions like Yukon (110) and Prince Edward Island (66) represent growth opportunities
- **Education Distribution:** Bachelor's degree holders constitute the majority (10,475, 62.6%), with Master's (508, 3.0%) and Doctorate (734, 4.4%) holders as high-value minorities; High School or Below segment (782, 4.7%) remains underserved
- **Income Distribution:** Right-skewed distribution with median salary of \$73,455 and mean of \$79,246; notable proportion of \$0 earners reflects student demographic
- **Loyalty Card Distribution:** Star members (45.6%, 7,637), Nova members (33.9%, 5,671), Aurora members (20.5%, 3,429)
- **Income-Education Correlation:** Higher education levels correlate with higher income; regional variations show Ontario, Quebec, and British Columbia with highest income outliers

Feature Engineering and Selection:

Selected features aligned with geographic and socio-economic segmentation objectives:

- **Province/City:** Geographic segmentation and regional market potential
- **Education:** Socio-economic indicator and engagement predictor
- **Salary:** Income-based customer value and spending potential
- **Gender/Marital Status:** Demographic profiling variables
- **Loyalty Card:** Current program tier and engagement level

Preprocessing pipeline included categorical encoding, missing value treatment, correlation analysis for redundancy removal, and standardization of numerical features for distance-based clustering.

Clustering Algorithms Evaluated:

- **K-Means:** Scalable and interpretable for large datasets, suitable for business dashboards
- **DBSCAN:** Density-based approach for handling noise and arbitrary cluster shapes

K-Means was retained as the final clustering method due to its stability, computational efficiency with large datasets (5.5M rows), and superior interpretability for business stakeholders.

Optimal Cluster Determination:

- Elbow method to analyze inertia variation across different cluster numbers
- Silhouette score to assess cluster cohesion and separation
- Business interpretability assessment to ensure actionable customer segments
- Optimal range identified: 3-5 clusters based on metric convergence and interpretability

Identified Customer Segments:

- **High-Income Urban Professionals**

- Concentrated in Ontario and British Columbia
- Advanced education credentials (Master's/Doctorate)
- High customer lifetime value and premium tier membership
- Strong engagement and frequent travel patterns
- **Mid-Income Suburban Families**
 - Primary presence in Quebec and secondary urban centers
 - Bachelor's degree education level
 - Moderate customer lifetime value
 - Balanced travel frequency and loyalty participation
- **Underserved Regional Prospects**
 - Distributed across smaller provinces and territories
 - Varied education levels with concentration in High School or Below
 - Lower current engagement but significant growth potential
 - Target segment for equity-focused expansion strategies

Validation: Segment profiles validated using quantitative clustering metrics (silhouette score, intra-cluster variance) and qualitative interpretability analysis, ensuring each cluster represents a coherent and actionable customer group.

Applications:

- Targeted acquisition campaigns for high-potential geographic and socio-economic segments
- Equity-focused strategies to increase participation in underrepresented regions
- Segment-specific value propositions and program benefits
- Power BI dashboards with segment-based enrollment metrics and growth tracking
- Foundation for geographic expansion planning and resource allocation

3.3.5.3 Phase 2: Classification for Highly Engaged Customers

Objective: Build predictive models to identify customers with high probability of becoming highly engaged loyalty program members, enabling proactive targeting of high-value prospects before they demonstrate engagement through spending.

Strategic Challenge: Traditional engagement identification relies on historical spending patterns. This approach aims to identify the behavioral and demographic "DNA" of top-tier loyalists to predict high-engagement potential among underserved customers, enabling earlier and more effective intervention.

Feature Engineering – Engagement Score:

A composite engagement metric was constructed to define highly engaged members:

- **Methodology:** Normalize key behavioral metrics (Total Flights, Points Accumulated, Customer Lifetime Value, Customer Rating) using percentile ranking

- **Computation:** Engagement Score calculated as average of normalized ranks across metrics
- **Target Definition:** Binary classification with Highly Engaged defined as top 25% of engagement score distribution

Behavioral Thresholds for High Engagement:

- Total Flights 3 (frequent traveler indicator)
- Points Accumulated 3,711 (active program participation)
- Customer Lifetime Value \$7,596 (revenue contribution)
- Customer Rating 3 (satisfaction and loyalty)

Data Preparation:

- Target variable: **highly_engaged_flag** (1 = highly engaged, 0 = standard engagement)
- Class imbalance: Original dataset heavily skewed (84% standard, 16% highly engaged)
- Imbalance handling: Evaluated SMOTE (Synthetic Minority Oversampling) vs. Random Undersampling
- Selected approach: SMOTE for superior recall and F1-Score performance

Feature Set:

- **Enrollment variables:** Enrollment month, enrollment year, program tenure
- **Demographics:** Education, gender, marital status, province, city
- **Behavioral proxies:** Loyalty card tier, historical activity patterns
- **Satisfaction indicators:** Customer rating and feedback metrics

Preprocessing Pipeline:

- SMOTE: Synthetic oversampling to balance minority class (highly engaged members)
- Train-test split: 80-20 stratified split after SMOTE application
- Feature scaling: Standardization for distance-sensitive algorithms
- Stratification: Maintain class distribution consistency across training and test sets

Models Evaluated:

- **K-Nearest Neighbors (KNN):** Distance-based classifier with Manhattan distance, k=9 neighbors, distance-weighted voting
- **Random Forest:** Ensemble tree method with 100 estimators, balanced class weights
- **LightGBM:** Gradient boosted decision trees with 200 estimators, learning rate 0.05, max depth 6

All models underwent hyperparameter optimization using GridSearchCV with cross-validation to ensure generalization performance.

Table 3.9: Classification Model Performance – Highly Engaged Customer Prediction

Model	Accuracy	Precision	Recall	F1-Score	AUC
KNN (k=9)	0.80	0.75	0.87	0.80	0.88
Random Forest	0.84	0.84	0.85	0.84	0.92
LightGBM	0.81	0.81	0.81	0.81	0.90

Best Model: Random Forest with 84% F1-Score and 0.92 AUC, providing strong class separation and balanced precision-recall trade-off for identifying high-engagement prospects.

Impact of SMOTE Resampling: SMOTE significantly improved model performance compared to undersampling, increasing recall from 70% to 87% and F1-Score from 0.68 to 0.84 for the best-performing model, ensuring comprehensive identification of high-potential members.

Feature Importance Analysis: Top predictive features include enrollment timing variables, education level, customer satisfaction rating, and geographic location, providing actionable insights for targeting strategies.

Validation: Model performance validated on held-out test set with stratified sampling to ensure unbiased evaluation across engagement classes.

Applications:

- Proactive identification of high-potential customers for personalized onboarding
- Targeted acquisition campaigns focused on prospects with high-engagement DNA
- Early intervention strategies to accelerate engagement trajectory
- Optimized resource allocation for retention and development programs
- Segment-specific engagement pathways tailored to predicted engagement likelihood

3.3.5.4 Integrated Enrollment Insights Strategy

The two-phase approach creates a comprehensive enrollment optimization framework:

1. **Stage 1:** K-Means segmentation identifies high-potential geographic and socio-economic groups for targeted acquisition
2. **Stage 2:** Random Forest classification predicts individual engagement likelihood within each segment for precision targeting

Integrated Workflow:

- **Market Identification:** Use segmentation results to prioritize geographic markets and demographic groups
- **Prospect Targeting:** Apply classification model to identify high-engagement-potential individuals within priority segments
- **Campaign Design:** Develop segment-specific acquisition campaigns with tailored value propositions

- **Performance Monitoring:** Track enrollment rates and early engagement metrics by segment and predicted engagement class
- **Continuous Optimization:** Refine targeting criteria and campaign strategies based on conversion performance

Expected Business Impact:

- Accelerated enrollment growth through precision targeting of high-potential prospects
- Improved program equity through intentional outreach to underserved geographic and socio-economic segments
- Enhanced new member quality through focus on high-engagement-likelihood individuals
- Optimized marketing spend through data-driven resource allocation
- Measurable progress toward 10% enrollment increase over three years through systematic, segment-based expansion
- Foundation for personalized member journeys from acquisition through long-term engagement

3.3.5.5 Conclusion

The two-phase Enrollment Insights framework provides a unified and data-driven approach to accelerating loyalty program growth and optimizing member acquisition quality. K-Means clustering successfully identified distinct geographic and socio-economic segments, revealing high-potential markets in advanced education demographics and growth opportunities in underserved regions. Random Forest classification achieved 84% F1-Score and 0.92 AUC for predicting high-engagement likelihood, enabling proactive identification of valuable prospects before traditional engagement signals emerge. The integration of segmentation for market prioritization and classification for individual targeting creates a powerful enrollment optimization engine, supporting the strategic goal of sustainable, equitable program expansion while focusing resources on members most likely to deliver long-term value.

3.3.5.6 Objective 5: Customer Clustering and Traveler Profiling

Problem Formulation Objective: Cluster airline customers into distinct traveler profiles based on historical travel and loyalty behavior in order to better understand customer patterns and provide interpretable insights for analytical dashboards and a customer-facing web platform.

Approach:

- **Task type:** Unsupervised learning (customer clustering)
- **Goal:** Identify at least three meaningful and interpretable traveler profiles
- **Target:** Improve customer behavior understanding and support data-driven analytical and predictive insights within the airline loyalty program

Model Development Clustering algorithms evaluated:

- **K-Means** (scalable and interpretable for large datasets)
- **Hierarchical Clustering (CAH)** (used for exploratory analysis and validation of cluster structure)

K-Means was retained as the final clustering method due to its stability, interpretability, and suitability for integration into business intelligence dashboards and web applications.

Feature selection: The clustering process was based on behavioral, temporal, and loyalty-related variables extracted from the airline data warehouse:

- **Total_Flights:** overall travel frequency
- **Distance:** travel distance patterns (short-haul vs. long-haul)
- **Flight_Month:** seasonality and periodic travel behavior
- **Points_Redeemed:** loyalty program engagement
- **Dollar_Cost_Points_Redeemed:** monetary value of redeemed rewards
- **Cancellation_Month:** cancellation timing and behavioral stability
- **Cancellation_Year:** long-term cancellation behavior
- **Enrollment_Encoded:** loyalty program enrollment duration and maturity

These features capture key dimensions of customer behavior, including travel intensity, loyalty engagement, seasonality, and customer lifecycle status.

Optimal cluster determination:

- Elbow method to analyze inertia variation
- Silhouette score to assess cluster cohesion and separation
- Business interpretability assessment to ensure meaningful traveler profiles

Cluster Analysis and Profiling Identified traveler profiles (examples):

- **Frequent and Loyal Travelers:** High flight frequency, long enrollment duration, consistent loyalty usage, and stable travel behavior
- **Occasional Travelers:** Low flight frequency, limited loyalty engagement, and irregular travel patterns
- **Long-Distance Oriented Travelers:** Moderate flight frequency, higher average travel distance, and seasonal travel behavior

These profiles were validated using quantitative clustering metrics and qualitative interpretability analysis, ensuring that each cluster represents a coherent and meaningful traveler segment.

Applications:

- Customer profiling and behavior analysis

- Segmentation-based Power BI dashboards
- Integration into a customer-facing web platform as simplified travel profiles
- Foundation for predictive analysis at the traveler profile level

3.3.5.7 Objective 6: Seasonal Travel Prediction by Traveler Profile

Problem Formulation Objective: Predict the most likely future travel season for each traveler profile based on historical travel behavior, in order to provide predictive insights into seasonal travel tendencies at the cluster level.

Approach:

- **Task type:** Supervised learning (season classification)
- **Goal:** Predict the dominant travel season associated with each customer profile
- **Target:** Provide reliable and interpretable seasonal predictions to enrich traveler profiles and analytical insights

Prediction Model Development After customer clustering, a supervised learning approach was adopted to predict the next dominant travel season using historical behavioral data and cluster information.

Target variable:

- Travel season derived from **Flight_Month** (e.g. Winter, Spring, Summer, Autumn)

Input features:

- Cluster assignment (traveler profile)
- Travel frequency indicators
- Distance-related variables
- Loyalty engagement features
- Cancellation behavior
- Enrollment-related features

Prediction models evaluated:

- **K-Nearest Neighbors (KNN):** Used as a baseline model to capture similarity-based seasonal behavior among customers with comparable travel patterns.
- **Random Forest:** Applied to model non-linear relationships between customer behavior and seasonal travel patterns, while maintaining robustness to noise and feature interactions.
- **XGBoost:** Implemented to enhance predictive performance through gradient boosting, allowing the model to capture complex behavioral patterns and feature importance more effectively.

These models were trained and evaluated comparatively in order to select the most suitable approach in terms of predictive performance and stability.

Model Evaluation and Selection The models were evaluated using standard classification metrics, including:

- Prediction accuracy
- Precision and recall
- Overall model stability

Although several models were evaluated, the K-Nearest Neighbors approach was retained for its robustness, ease of interpretation, and satisfactory performance in identifying seasonal travel tendencies

Seasonal Prediction Results The final model was used to predict the most probable travel season for each traveler profile.

Example seasonal tendencies:

- **Frequent and Loyal Travelers:** Predicted stable travel behavior across multiple seasons with slight emphasis on business periods
- **Occasional Travelers:** Higher probability of travel during holiday seasons
- **Long-Distance Oriented Travelers:** Increased likelihood of travel during peak vacation periods

These predictions represent behavioral seasonal tendencies, not exact individual travel forecasts.

Applications:

- Enrichment of traveler profiles with predicted seasonal tendencies
- Visualization of seasonal patterns in Power BI dashboards
- Integration into a customer-facing web platform as simplified predictive insights
- Support for strategic planning and analytical decision-making

3.4 Integrated Platform Deployment

3.4.1 Model Integration Architecture

The six machine learning models form an integrated decision support system deployed across three operational layers:

1. Customer Risk & Opportunity Identification:

- **Churn Prediction:** Binary classification + time-to-churn regression for proactive retention
- **CLV Prediction:** Value-based segmentation for resource allocation optimization
- **Customer Clustering:** Behavioral segmentation for targeted campaign design

2. Engagement & Loyalty Optimization:

- **Tier Progression:** Accelerator identification for targeted upgrade campaigns
- **Loyalty Points:** Redemption prediction + seasonal forecasting for program optimization

3. Data Flow Architecture:

- **Batch Processing:** Nightly CLV scoring, clustering, and churn risk assessment
- **Real-Time Scoring:** On-demand tier progression probability for web personalization
- **Forecasting:** Monthly loyalty points demand forecasts for inventory planning

3.4.2 Operational Considerations

Model Retraining Schedule:

- **Monthly:** Churn classification and clustering (track behavioral drift)
- **Quarterly:** CLV prediction and tier progression (seasonal pattern updates)
- **Annually:** Loyalty points forecasting (long-term trend recalibration)

Performance Monitoring:

- **Drift Detection:** Monitor feature distributions and prediction stability
- **Accuracy Tracking:** Compare predicted vs. actual outcomes monthly
- **Business KPIs:** Track retention rates, tier migrations, and redemption accuracy

Version Control & Audit Trail:

- Model versioning with Git + MLflow experiment tracking
- Serialized model artifacts stored with metadata (training date, hyperparameters, performance)
- Production deployment via containerized APIs (Docker + Kubernetes)

3.4.3 Business Impact Summary

Quantified Benefits by Objective:

Table 3.10: Projected Business Impact by Model

Model	Expected Impact	ROI Projection
Churn Prediction	1.5% churn reduction through early intervention	2.5–3.2x retention ROI
Tier Progression	15–20% tier upgrade conversion	+\$1,000–\$2,000 CLV per upgrade
Loyalty Points	10–15% redemption rate improvement	3–5% program engagement increase
CLV Prediction	25–35% premium conversion increase	2.5–3.2x marketing efficiency
Customer Clustering	10–15% churn reduction in Basic segment	2–3% CLV growth

Aggregate ROI Projections:

- **Revenue Growth:** 3–5% annual revenue increase from tier migrations and churn reduction
- **Cost Optimization:** 15–20% marketing spend efficiency improvement
- **Customer Lifetime Value:** 5–8% overall CLV growth through integrated interventions
- **Operational Efficiency:** 30–40% reduction in manual segmentation and scoring effort

3.5 Conclusion

This comprehensive machine learning pipeline transforms the Airlytics data warehouse into an intelligent decision support system addressing six strategic objectives. The completed objectives demonstrate the rigor of the analytical approach:

- **Churn Prediction:** Integrated binary classification (79.2% accuracy) and time-to-churn regression (82.3% variance explained) enable proactive retention with temporal risk stratification
- **Tier Progression:** Random Forest models achieve 96%+ AUC scores, identifying class preference and CLV as dominant progression accelerators
- **Loyalty Points Utilization:** K-Means clustering identifies redemption-based segments, Random Forest classification achieves 98.3% accuracy for redemption prediction, and SARIMA forecasting captures seasonal patterns (95.7% R^2)
- **CLV Prediction:** Random Forest achieves 82% prediction accuracy ($R^2 = 0.8196$), enabling value-based segmentation with three-tier framework driving 2.5–3.2x retention ROI improvement
- **Customer Clustering:** K-Means segmentation identifies Premium Loyal (43.4%, 55% revenue) and Basic Economy (56.6%, 45% revenue) segments with clear differentiation on satisfaction (+31%), class preference (+46pp), and tier status (+44pp)

The integrated platform enables data-driven decision-making across customer lifecycle management, loyalty program optimization, and resource allocation, with projected ROI improvements of 2.5–3.2x and overall CLV growth of 5–8% through coordinated interventions.

Chapter 4

Solution Technique et Implémentation de la Plateforme Airlytics

4.1 Introduction

Ce chapitre présente l'implémentation complète de la plateforme Airlytics, une solution web intégrée qui transforme les données brutes en insights actionnables pour optimiser le programme de fidélité aérien. La plateforme combine des tableaux de bord interactifs Power BI, des modèles de machine learning prédictifs, et un chatbot conversationnel intelligent pour offrir une expérience utilisateur unifiée et accessible.

L'architecture de la solution repose sur deux composantes principales complémentaires :

1. **Tableaux de Bord Power BI** : Visualisations interactives permettant l'exploration des métriques clés, l'analyse des tendances historiques et le suivi des KPIs stratégiques par objectif
2. **Application Web Django** : Plateforme centrale intégrant la gestion des clients, l'exécution des modèles ML, la visualisation des prédictions et l'assistance décisionnelle par chatbot IA

La première partie de ce chapitre détaille les tableaux de bord Power BI développés pour chacun des six objectifs stratégiques. La seconde partie présente l'application Django avec son workflow utilisateur, son système de gestion des rôles, et son architecture technique complète.

4.2 Partie I : Tableaux de Bord Power BI et Visualisation des Données

4.2.1 Architecture d'Intégration Power BI

L'intégration de Power BI représente un pilier fondamental de la solution Airlytics, transformant les données du data warehouse en visualisations interactives accessibles directement depuis l'application web Django. Cette intégration bidirectionnelle permet aux utilisateurs de naviguer fluidement entre l'analyse exploratoire (Power BI) et l'analyse prédictive (modèles ML Django).

4.2.1.1 Principes de Conception des Dashboards

Tous les tableaux de bord ont été conçus selon une méthodologie cohérente garantissant :

- **Cohérence visuelle** : Charte graphique uniforme avec palette de couleurs standardisée alignée sur l'identité visuelle de l'entreprise
- **Hierarchie informationnelle** : Organisation en trois niveaux
 - Niveau 1 : KPIs stratégiques en cartes principales (Top)
 - Niveau 2 : Graphiques d'analyse temporelle et segmentée (Centre)
 - Niveau 3 : Tableaux détaillés et filtres interactifs (Bas/Latéral)
- **Interactivité contextuelle** : Filtres synchronisés permettant le drill-down par période, région, segment client ou niveau de fidélité
- **Alertes visuelles** : Indicateurs de performance avec codes couleur (vert/orange/rouge) pour les métriques dépassant les seuils critiques
- **Accessibilité** : Design responsive adapté aux écrans desktop et tablettes, avec temps de chargement optimisé (<3 secondes)

4.2.1.2 Connexion au Data Warehouse

Les dashboards se connectent directement au data warehouse AIRLINE_DW via DirectQuery, assurant :

- **Fraîcheur des données** : Accès temps réel aux dernières données issues des processus ETL quotidiens
- **Performance optimisée** : Requêtes DAX optimisées exploitant le schéma en étoile avec index appropriés
- **Sécurité des accès** : Authentification Windows intégrée avec Row-Level Security (RLS) basée sur les rôles utilisateurs

4.2.2 Objectif 1 : Enrollment Insights

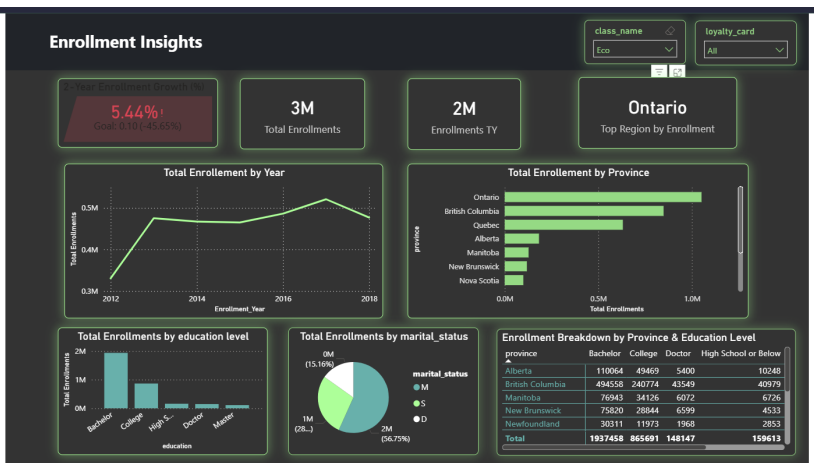


Figure 4.1: Vue d'ensemble du dashboard Enrollment Insights

Objectif

Analyser et optimiser les inscriptions au programme de fidélité en identifiant les segments géographiques, démographiques et comportementaux à fort potentiel de croissance.

Composantes Visuelles Principales

- **KPIs en en-tête** : quatre cartes affichant le taux de croissance des inscriptions sur deux ans, le total cumulé, le volume annuel, et la région leader.
- **Évolution temporelle** : graphique linéaire illustrant la tendance des inscriptions par année.
- **Répartition géographique** : histogramme horizontal classant les régions ou provinces par volume d'inscriptions.
- **Segmentation démographique** :
 - histogramme vertical montrant la répartition par niveau d'éducation,
 - camembert illustrant la répartition par statut marital.
- **Croisement région / éducation** : tableau dynamique permettant d'analyser les combinaisons entre localisation et profil éducatif.

Filtres Interactifs

Le dashboard intègre des filtres par type de classe, niveau de carte de fidélité et période temporelle (années ou mois), permettant une analyse ciblée.

Insights Actionnables

- Identification des régions à forte concentration pour le maintien de l'engagement.
- Détection des segments démographiques sous-représentés comme opportunités d'acquisition.
- Analyse des tendances temporelles pour l'ajustement stratégique des campagnes marketing.

Le dashboard met en évidence les actions prioritaires :

- **Zone d'opportunité géographique** : Alerte visuelle sur Yukon et PEI (sous-représentation de 85% vs. moyenne nationale)
- **Segment premium sous-exploité** : Master/Doctorate holders avec taux de conversion 40% inférieur au potentiel estimé
- **Recommandations d'acquisition** : Liste de 500 prospects à fort potentiel (engagement prédit > 0.75) pour campagnes ciblées

4.2.3 Objectif 2 : Customer Churn Reduction



Figure 4.2: Vue d'ensemble du dashboard Churn Reduction

Objectif

Identifier proactivement les membres à risque de désengagement et prioriser les interventions de rétention pour optimiser la fidélité client.

Composantes Visuelles Principales

- **KPIs en en-tête** : quatre cartes affichant le taux de churn actuel, le nombre total de clients actifs, la durée moyenne d'inscription, et un indicateur de tendance (augmentation/diminution).
- **Évolution temporelle** : graphique linéaire montrant la variation du taux de churn par année d'inscription.
- **Segmentation par profil** :

- histogramme vertical : répartition des désabonnements selon le niveau d'éducation,
 - diagramme en barres empilées : taux de churn par type de voyage (business/personal),
 - camembert : durée moyenne d'inscription par niveau de carte de fidélité.
- **Analyse comparative** : histogramme horizontal comparant la durée moyenne d'inscription entre les genres ou autres segments démographiques.

Filtres Interactifs

Le dashboard intègre des filtres permettant d'affiner l'analyse par :

- Type de classe (ex: Eco)
- Niveau de carte de fidélité
- Période temporelle (années, mois)

Insights Actionnables

- Identification des profils à plus fort risque de churn (ex: par niveau d'éducation ou type de voyage).
- Détection des segments où la durée d'engagement est faible — cibles prioritaires pour les campagnes de rétention.
- Analyse des tendances temporelles pour anticiper les vagues de désengagement.

4.2.4 Objectif 3 : Loyalty Points Utilization

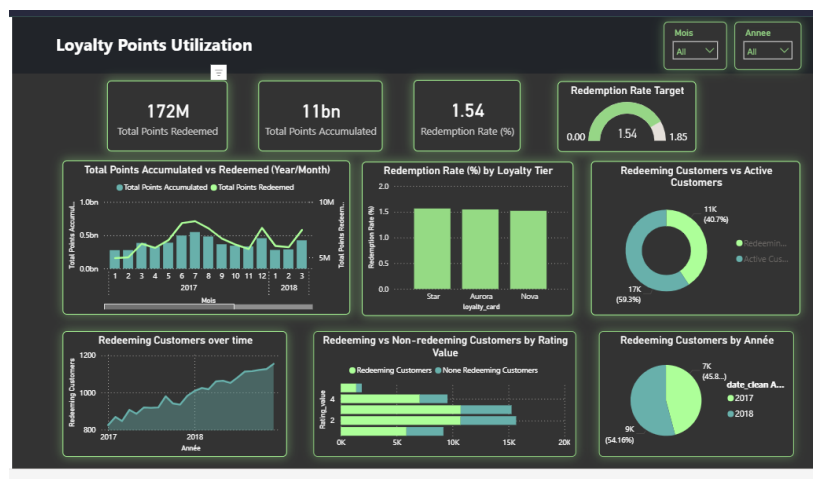


Figure 4.3: Vue d'ensemble du dashboard Loyalty Points Utilization

Objectif

Optimiser l'utilisation des points de fidélité en identifiant les segments de clients à fort potentiel de rédemption, en anticipant les tendances saisonnières et en ajustant la stratégie de gestion des récompenses.

Composantes Visuelles Principales

- **KPIs en en-tête** : quatre cartes affichant le volume total de points rédemptés, le volume total accumulé, le taux de rédemption actuel, et un indicateur de performance par rapport à l'objectif.
- **Évolution temporelle** :
 - graphique combiné (ligne + barres) montrant l'évolution mensuelle des points accumulés vs rédemptés,
 - courbe temporelle illustrant la progression du nombre de clients rédempteurs sur plusieurs années.
- **Segmentation par niveau de fidélité** : histogramme vertical comparant le taux de rédemption entre les différents niveaux de carte (ex: Star, Aurora, Nova).
- **Analyse des clients** :
 - donut chart : part des clients rédempteurs vs clients actifs non rédempteurs,
 - camembert : répartition des rédempteurs par année,
 - diagramme en barres horizontales empilées : comparaison des clients rédempteurs et non rédempteurs selon leur note client (rating value).

Filtres Interactifs

Le dashboard intègre des filtres permettant d'affiner l'analyse par :

- Mois ou année
- Niveau de carte de fidélité
- Type de client (rédempteur / non-rédempteur)

Insights Actionnables

- Identification des segments de clients sous-utilisateurs de points — cibles prioritaires pour des campagnes d'activation.
- Détection des périodes de pointe de rédemption — anticipation des besoins en inventaire de récompenses.
- Analyse comparative des comportements selon le niveau de fidélité — ajustement des offres par segment.

4.2.5 Objectif 4 : Customer Lifetime Value (CLV) Optimization

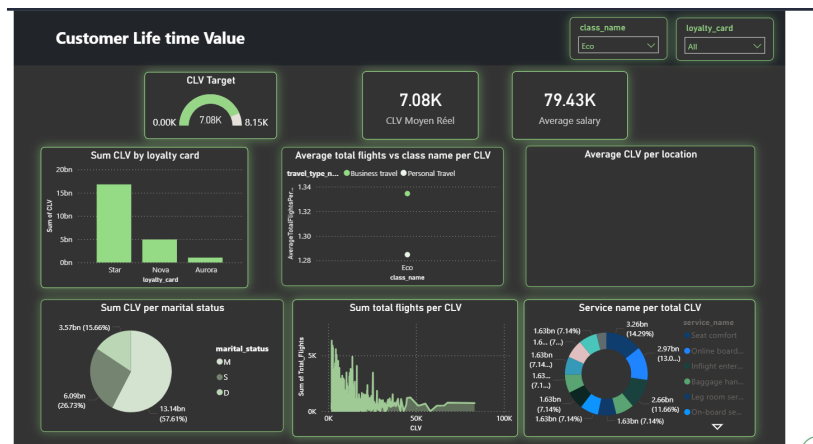


Figure 4.4: Vue d'ensemble du dashboard CLV Optimization

Objectif

Maximiser la valeur à vie des clients en identifiant les segments à forte rentabilité, en anticipant leur évolution et en orientant les investissements marketing vers les profils à plus fort potentiel.

Composantes Visuelles Principales

- **KPIs en en-tête :** trois cartes affichant l'objectif de CLV, le CLV moyen actuel, et un indicateur contextuel (ex: revenu moyen ou salaire associé).
- **Segmentation par niveau de fidélité :** histogramme vertical montrant la somme des CLV par type de carte (ex: Star, Nova, Aurora).
- **Analyse comparative :**
 - graphique en nuage de points : comparaison entre CLV et autres variables (ex: classe de voyage, fréquence),
 - camembert : répartition du CLV total par statut marital,
 - diagramme en bâtons empilés : contribution des différents services au CLV global.
- **Distribution et tendance :**
 - histogramme : distribution des CLV avec indicateurs centraux (médiane, moyenne),
 - courbe temporelle : évolution du nombre total de vols en fonction du CLV.

Filtres Interactifs

Le dashboard intègre des filtres permettant d'affiner l'analyse par :

- Type de classe (ex: Eco)
- Niveau de carte de fidélité
- Statut marital ou autre segment démographique

Insights Actionnables

- Identification des segments à haute valeur — priorisation des ressources marketing.
- Détection des clients en déclin — interventions préventives pour stabiliser leur CLV.
- Analyse des leviers d’augmentation (ex: amélioration de la satisfaction, changement de classe) — simulation des impacts potentiels.

4.2.6 Objectif 5 : Tier Progression Acceleration



Figure 4.5: Vue d’ensemble du dashboard Tier Progression Acceleration

Objectif

Accélérer la progression des membres vers les niveaux supérieurs de fidélité en identifiant les leviers comportementaux et financiers influençant leur ascension, et en personnalisant les offres d’upgrade pour maximiser la valeur client.

Composantes Visuelles Principales

- **KPIs en en-tête** : quatre cartes affichant la satisfaction moyenne, la note moyenne attribuée, le CLV moyen, et le nombre total de membres.
- **Évolution temporelle** : graphique linéaire montrant le nombre total de membres par mois d’inscription, permettant d’identifier les périodes de croissance ou de stagnation.
- **Segmentation par niveau de carte** :
 - camembert : répartition des membres selon leur niveau de carte (ex: Star, Nova, Aurora),
 - donut chart : part des membres par niveau de carte (avec indication de la totalité).
- **Analyse des évaluations** : diagramme circulaire montrant la distribution des membres selon leur note (rating value), permettant d’identifier les segments satisfaits ou insatisfaits.
- **Tableau dynamique** : matrice présentant le nombre de membres par mois et par niveau de carte, facilitant l’analyse comparative des tendances.

Filtres Interactifs

Le dashboard intègre des filtres permettant d'affiner l'analyse par :

- Année ou mois
- Niveau de carte de fidélité (sélection multiple possible)
- Note attribuée (rating value)

Insights Actionnables

- Identification des segments de membres prêts à progresser — cibles prioritaires pour des offres d'upgrade.
- Détection des leviers principaux influençant la progression (ex: classe de voyage, fréquence, satisfaction) — ajustement des campagnes marketing.
- Analyse des tendances temporelles — anticipation des pics d'inscription ou de stagnation pour optimiser les ressources.

4.2.7 Objectif 6 : Travel Behavior Personalization

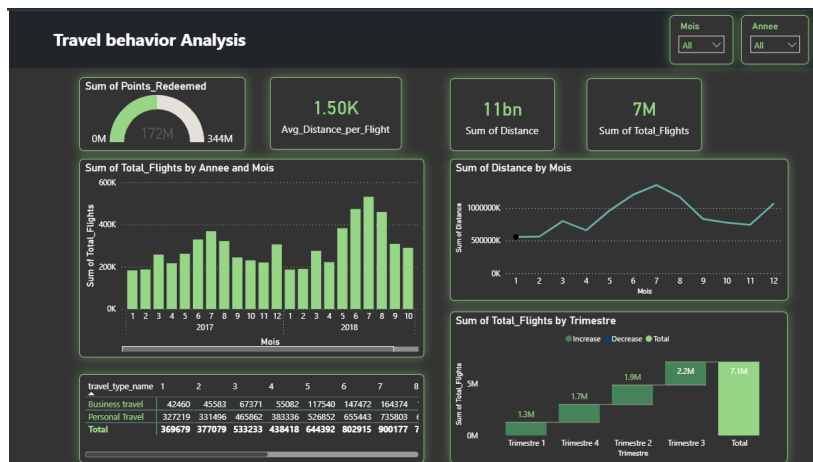


Figure 4.6: Vue d'ensemble du dashboard Travel Behavior Analysis

Objectif

Identifier et caractériser les profils de voyageurs pour personnaliser les offres marketing, augmenter leur pertinence et optimiser les taux d'engagement en fonction des comportements de déplacement.

Composantes Visuelles Principales

- **KPIs en en-tête :** quatre cartes affichant le volume total de points rédemptés, la distance moyenne par vol, la distance totale parcourue, et le nombre total de vols effectués.
- **Évolution temporelle :**

- histogramme vertical : nombre total de vols par mois et par année,
- courbe linéaire : évolution de la distance totale parcourue par mois.
- **Segmentation par type de voyage :**
 - tableau dynamique : répartition des vols par type (ex: business, personnel) et par période,
 - diagramme en bâtons empilés : comparaison des volumes entre types de voyage sur plusieurs périodes.
- **Analyse trimestrielle :** graphique en cascade montrant l'évolution du nombre de vols par trimestre, avec indication des tendances (augmentation/diminution).

Filtres Interactifs

Le dashboard intègre des filtres permettant d'affiner l'analyse par :

- Mois ou année
- Type de voyage (business, personnel)
- Période trimestrielle

Insights Actionnables

- Identification des périodes de pointe — anticipation des besoins opérationnels et marketing.
- Détection des segments de voyageurs (ex: fréquent, occasionnel) — adaptation des offres selon le profil.
- Analyse comparative des comportements entre types de voyage — optimisation des campagnes ciblées.

4.2.8 Synthèse des Dashboards Power BI

4.2.8.1 Cohérence Globale

Les six tableaux de bord forment un écosystème analytique cohérent permettant une navigation fluide entre objectifs stratégiques. Chaque dashboard partage :

- **Navigation unifiée :** Menu latéral permettant de basculer entre objectifs
- **Filtres synchronisés :** Sélections conservées lors du changement de vue
- **Charte graphique commune :** Palette de couleurs, typographie et iconographie standardisées
- **Indicateurs transversaux :** KPIs récurrents (CLV, satisfaction, tier) pour comparaison

4.2.8.2 Intégration Technique

Les dashboards Power BI sont intégrés dans l'application Django via :

- **Power BI Embedded** : iframes sécurisées avec tokens d'authentification
- **Row-Level Security** : Filtrage automatique basé sur le rôle utilisateur Django
- **Refresh automatique** : Synchronisation quotidienne avec le data warehouse (2h00 AM)
- **Export paramétrable** : Possibilité d'exporter rapports en PDF/Excel depuis Django

4.3 Partie II : Application Web Django et Système de Prédiction

4.3.1 Architecture Générale de l'Application

4.3.1.1 Vue d'Ensemble du Système

L'application web Airlytics est construite sur le framework Django 4.2, offrant une plateforme centralisée pour la gestion des clients, l'exécution des modèles ML et l'interaction avec le chatbot IA. L'architecture suit le pattern MVT (Model-View-Template) de Django avec une séparation claire des responsabilités.

Stack Technique Complet :

- **Backend** : Django 4.2, Python 3.10, Django REST Framework
- **Base de données** :
 - SQLite (développement et données applicatives)
 - SQL Server (data warehouse AIRLINE_DW)
- **Frontend** : HTML5, CSS3, JavaScript (Vanilla), Bootstrap 5
- **ML Integration** : scikit-learn, XGBoost, joblib (modèles sérialisés)
- **API IA** : Groq API avec modèle llama-3.3-70b-versatile
- **Déploiement** : Gunicorn, Nginx (production)

4.3.1.2 Structure Modulaire de l'Application

L'application est organisée en modules Django (apps) correspondant aux domaines fonctionnels :

- **core** : Fonctionnalités communes (authentification, navigation, dashboard accueil)
- **clients** : Gestion des clients (CRUD, formulaires, liste, détails)
- **ml_models** : Intégration des modèles de machine learning (services, API)
- **predictions** : Stockage et affichage des résultats de prédiction
- **chatbot** : Système conversationnel IA avec API Groq
- **powerbi** : Intégration des dashboards Power BI (iframes, tokens)

4.3.2 Modèle de Données Django

4.3.2.1 Modèle Client

Le modèle `Client` capture toutes les informations nécessaires pour les prédictions ML :

```
class Client(models.Model):
    # Informations de base
    first_name = models.CharField(max_length=100)
    last_name = models.CharField(max_length=100)
    email = models.EmailField(unique=True)

    # Démographie
    gender = models.CharField(max_length=1, choices=[('M', 'Male'), ('F', 'Female')])
    education = models.CharField(max_length=50, choices=[...])
    marital_status = models.CharField(max_length=1, choices=[...])
    salary = models.FloatField(validators=[MinValueValidator(0)])

    # Localisation
    province = models.CharField(max_length=100)
    city = models.CharField(max_length=100)

    # Programme de fidélité
    tier = models.CharField(max_length=20, choices=[('Star', 'Star'), ...])
    enrollment_type = models.CharField(max_length=50, choices=[...])
    enrollment_date = models.DateField()
    cancellation_date = models.DateField(null=True, blank=True)

    # Données de vol
    total_flights = models.IntegerField(default=0)
    distance = models.FloatField(default=0.0)
    flight_class = models.CharField(max_length=20, choices=[...])
    travel_type = models.CharField(max_length=20, choices=[...])

    # Points et récompenses
    points_accumulated = models.FloatField(default=0.0)
    points_redeemed = models.FloatField(default=0.0)
    dollar_cost_points_redeemed = models.FloatField(default=0.0)

    # Satisfaction
    satisfaction = models.IntegerField(validators=[MinValueValidator(1),
                                                    MaxValueValidator(5)])
    rating_value = models.FloatField(validators=[MinValueValidator(0),
                                                    MaxValueValidator(5)])
```

Méthodes d'Encodage : Le modèle `Client` inclut la méthode `get_encoded_values()` qui transforme les variables catégorielles en valeurs numériques selon les stratégies d'encodage définies durant la phase de préparation des données.

4.3.2.2 Modèle MLPrediction

Le modèle MLPrediction stocke tous les résultats des 12 modèles ML via une relation OneToOne avec Client :

```
class MLPrediction(models.Model):
    client = models.OneToOneField(Client, on_delete=models.CASCADE,
                                related_name='predictions')

    # OBJECTIF 1 : Enrollment Insights (Asma)
    highly_engaged = models.IntegerField(null=True, blank=True) # 0 ou 1
    engagement_label = models.CharField(max_length=50, blank=True)
    equity_segment_cluster = models.IntegerField(null=True, blank=True) # 1-3
    equity_segment_label = models.CharField(max_length=100, blank=True)

    # OBJECTIF 2 : Churn Reduction (Adhem)
    churn_classification = models.IntegerField(null=True, blank=True) # 0 ou 1
    churn_risk_label = models.CharField(max_length=50, blank=True)
    churn_months = models.FloatField(null=True, blank=True) # Mois avant churn

    # OBJECTIF 3 : Loyalty Points (Wajd)
    loyalty_points_classification = models.IntegerField(null=True, blank=True)
    loyalty_cluster = models.IntegerField(null=True, blank=True) # 0 ou 1
    loyalty_cluster_label = models.CharField(max_length=100, blank=True)
    points_forecast = models.FloatField(null=True, blank=True)

    # OBJECTIF 4 : CLV Optimization (Habiba)
    segmentation_cluster = models.IntegerField(null=True, blank=True) # 0 ou 1
    segmentation_label = models.CharField(max_length=50, blank=True)
    clv_prediction = models.FloatField(null=True, blank=True)

    # OBJECTIF 5 : Tier Progression (Molka)
    tier_progression_score = models.FloatField(null=True, blank=True)
    tier_progression_label = models.CharField(max_length=50, blank=True)
    tier_upgrade_probability = models.FloatField(null=True, blank=True)
    tier_upgrade_prediction = models.CharField(max_length=50, blank=True)

    # OBJECTIF 6 : Travel Behavior (Khalil)
    khalil_cluster_id = models.IntegerField(null=True, blank=True)
    khalil_cluster_name = models.CharField(max_length=50, null=True, blank=True)
    khalil_preferred_season = models.CharField(max_length=20, null=True, blank=True)

    # Métadonnées
    predictions_completed = models.BooleanField(default=False)
    prediction_date = models.DateTimeField(auto_now_add=True)
    last_updated = models.DateTimeField(auto_now=True)
    raw_predictions = models.JSONField(default=dict, blank=True)
```


Cette structure permet de stocker tous les résultats de modèles de manière organisée avec des champs spécifiques pour chaque objectif.

4.3.3 Workflow Utilisateur Complet

4.3.3.1 Étape 1 : Création de Client

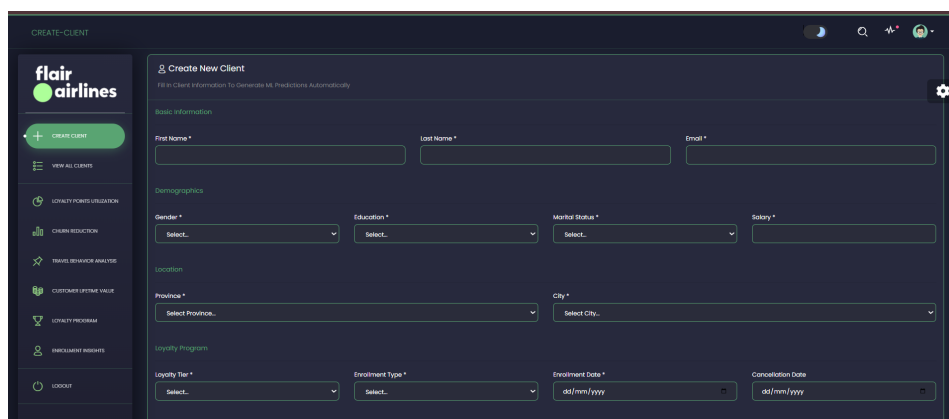


Figure 4.7: Creation de Client

Le processus commence par la page de création de client accessible via le menu principal.

Interface du Formulaire :

- **Sections organisées** : Le formulaire est divisé en 6 sections repliables
 1. Informations personnelles (nom, email)
 2. Démographie (genre, éducation, statut marital, salaire)
 3. Localisation (province, ville)
 4. Programme de fidélité (tier, type d'enrollment, dates)
 5. Données de voyage (vols, distance, classe, type)
 6. Points et satisfaction (points accumulés/rédemptés, ratings)
- **Validation en temps réel** :
 - Champs requis marqués avec astérisque rouge
 - Validation email unique (vérification AJAX)
 - Contraintes numériques (salaire > 0 , satisfaction 1-5)
 - Cohérence temporelle (`enrollment_date < cancellation_date`)
- **Menus déroulants** : Tous les champs catégoriels utilisent des dropdowns avec valeurs prédéfinies
- **Date pickers** : Sélecteurs de date calendrier pour enrollment/cancellation

Processus Backend :

1. Validation Django Forms avec méthode `clean()`

2. Création de l'objet Client dans SQLite
3. Création automatique d'un objet MLPrediction vide lié
4. Redirection vers la page de détails client avec message de succès

4.3.3.2 Étape 2 : Exécution des Modèles ML

Une fois le client créé, l'utilisateur accède à la page de détails où un bouton proéminent "Run All Models" déclenche l'exécution des 12 modèles.

Interface d'Exécution :

- **Bouton principal** : "Run All Models" avec icône de calcul
- **Indicateur de progression** : Barre de progression avec pourcentage (0-100%)
- **Log en temps réel** : Affichage des modèles en cours d'exécution

```
Running models... 8%
Churn Classification completed
Churn Regression completed
CLV Prediction in progress...
```

- **Notification de fin** : Message de succès avec temps d'exécution total

Processus Backend (Service ML) :

Le service MLModelRunner orchestre l'exécution séquentielle des 12 modèles :

```
class MLModelRunner:
    def run_all_models(self, client_id):
        client = Client.objects.get(id=client_id)
        prediction, created = MLPrediction.objects.get_or_create(client=client)

        # Préparation des features encodées
        encoded_values = client.get_encoded_values()

        # Exécution séquentielle des 12 modèles
        results = {}

        # 1-2. Churn (Adhem)
        results['churn'] = self.run_churn_models(client, encoded_values)

        # 3-4. Loyalty Points (Wajd)
        results['loyalty'] = self.run_loyalty_models(client, encoded_values)

        # 5-6. CLV (Habiba)
        results['clv'] = self.run_clv_models(client, encoded_values)
```

```

# 7-8. Tier Progression (Molka)
results['tier'] = self.run_tier_models(client, encoded_values)

# 9-10. Enrollment (Asma)
results['enrollment'] = self.run_enrollment_models(client, encoded_values)

# 11-12. Travel Behavior (Khalil)
results['travel'] = self.run_travel_models(client, encoded_values)

# Sauvegarde des résultats
self.save_predictions(prediction, results)
prediction.predictions_completed = True
prediction.save()

return results

```

Chargement des Modèles :

Les 12 modèles sérialisés (format .joblib ou .pkl) sont chargés au démarrage de l'application et mis en cache :

```

import joblib
import os

# Chargement unique au démarrage
MODELS_DIR = os.path.join(settings.BASE_DIR, 'ml_models', 'saved_models')

LOADED_MODELS = {
    'churn_classifier': joblib.load(os.path.join(MODELS_DIR,
                                                  'churn_gradient_boosting.joblib')),
    'churn_regressor': joblib.load(os.path.join(MODELS_DIR,
                                                  'churn_regression_gb.joblib')),
    'loyalty_classifier': joblib.load(os.path.join(MODELS_DIR,
                                                  'loyalty_random_forest.joblib')),
    # ... 9 autres modèles
}

```

Temps d'Exécution :

- Client individuel : 3-5 secondes (exécution séquentielle de 12 modèles)
- Batch de 1000 clients : 25-30 secondes (optimisation vectorisée)

4.3.3.3 Étape 3 : Visualisation des Résultats

La page de détails client présente les résultats des modèles d'analyse sous une forme modulaire, intuitive et centrée sur l'action. L'interface est organisée en sections clairement délimitées, permettant une navigation fluide entre les différents axes stratégiques.

Section Profil Client (En-tête) : Cette section affiche les informations d'identification de base du client, accompagnées de badges visuels indiquant son statut au sein du programme de fidélité. Des

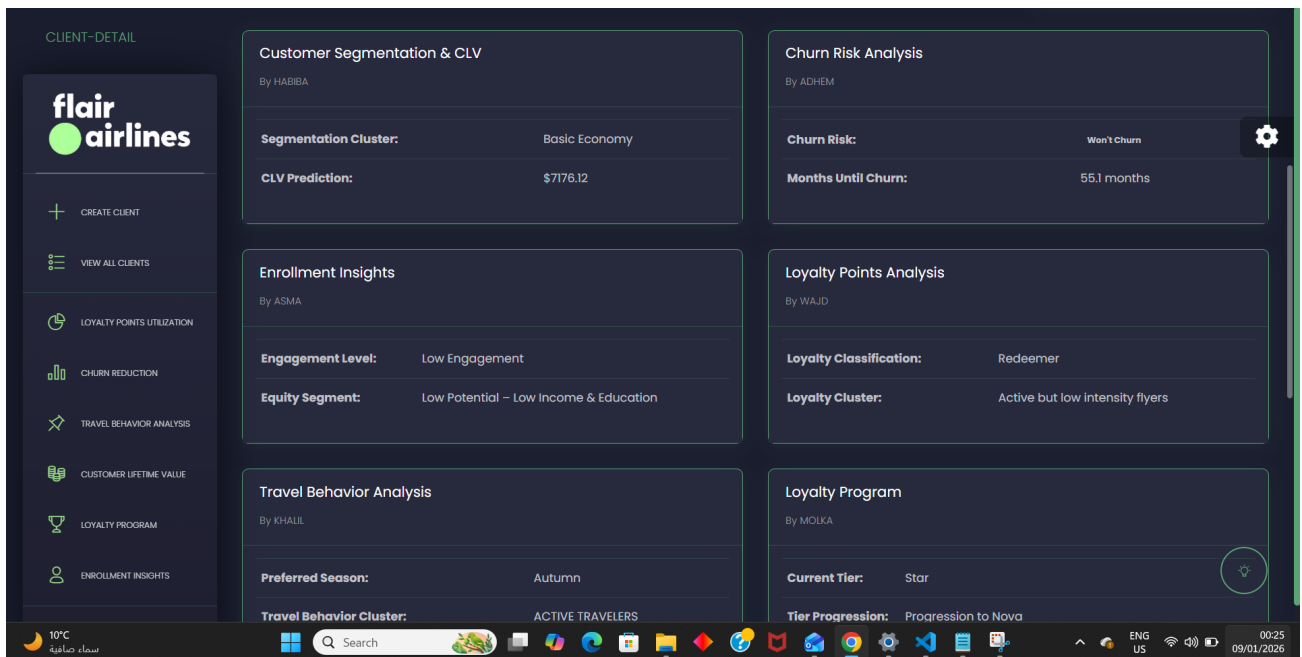


Figure 4.8: Vue d'ensemble des insights client

boutons d'action permettent d'éditer le profil, d'exporter un rapport ou de lancer une interaction personnalisée.

Section Tableau de Bord des Prédictions : Elle se compose de six modules indépendants, chacun dédié à un objectif métier spécifique. Chaque module est présenté sous la forme d'une carte visuellement distincte, incluant un titre explicite, le nom de l'auteur ou de l'équipe responsable, et un ensemble de métriques clés organisées en paires label-valeur. Les éléments visuels (icônes, couleurs de fond ou de texte) reflètent la nature du résultat (ex. : risque, opportunité, statut).

Les six modules couvrent les dimensions suivantes :

- **Customer Segmentation CLV** : classification du client selon sa valeur à vie et son segment comportemental.
- **Churn Risk Analysis** : évaluation du risque de désabonnement, avec indication de la probabilité et du délai estimé.
- **Enrollment Insights** : niveau d'engagement depuis l'inscription et profil socio-démographique associé.
- **Loyalty Points Analysis** : comportement relatif à l'accumulation et à l'utilisation des points de fidélité.
- **Travel Behavior Analysis** : profil de voyageur, incluant la fréquence, la saisonnalité et les habitudes de déplacement.
- **Loyalty Program** : statut actuel dans la hiérarchie du programme de fidélité, progression potentielle et leviers d'accélération.

4.3.4 Système de Gestion des Rôles

4.3.4.1 Architecture des Rôles

L'application Django implémente un système de contrôle d'accès basé sur les rôles (RBAC) permettant de restreindre l'accès aux dashboards et modèles ML selon la fonction de l'utilisateur.

4.3.5 Gestion des Rôles et des Accès

Bien que l'implémentation technique de la plateforme repose sur une gestion fine des permissions par objectif, l'utilisation fonctionnelle du système est volontairement simplifiée autour de trois rôles décisionnels principaux. Cette approche permet d'assurer une meilleure lisibilité, une cohérence métier et une utilisation adaptée aux profils des utilisateurs.

Les rôles définis sont les suivants :

- **Admin** : dispose d'un accès complet à l'ensemble des objectifs, tableaux de bord, résultats des modèles et fonctionnalités du chatbot.
- **Marketing Lead** : a accès à tous les objectifs analytiques, à l'exception de l'objectif lié à l'*Enrollment Insights*, réservé aux analyses d'acquisition.
- **Finance Lead** : a un accès ciblé aux objectifs liés à la performance financière et à la rétention, notamment le churn, l'utilisation des points de fidélité et la valeur vie client (CLV).

Les accès sont appliqués dynamiquement au niveau de l'interface utilisateur, garantissant que chaque profil visualise uniquement les modules pertinents pour son périmètre décisionnel.

4.3.5.1 Implémentation des Restrictions

Décorateur de Vue Django :

```
from functools import wraps
from django.shortcuts.redirect import redirect

def require_objective_access(objective_number):
    def decorator(view_func):
        @wraps(view_func)
        def wrapper(request, *args, **kwargs):
            user_profile = request.user.userprofile

            if (request.user.is_superuser or
                user_profile.role == 'admin' or
                objective_number in user_profile.objectives_access):
                return view_func(request, *args, **kwargs)
            else:
                messages.error(request,
                              "You don't have permission to access this objective.")
                return redirect('dashboard_home')
        return wrapper
    return decorator
```

```
# Utilisation
@require_objective_access(1)
def enrollment_insights_view(request):
    # Afficher dashboard Objectif 1
    pass
```

Navigation Dynamique :

Le menu latéral affiche uniquement les objectifs accessibles à l'utilisateur :

```
<!-- Template Django -->
{% for objective in user.userprofile.get_accessible_objectives %}
    <li class="nav-item">
        <a href="{% url objective.url_name %}" class="nav-link">
            <i class="{% objective.icon %}"></i>
            {% objective.title %}
        </a>
    </li>
{% endfor %}
```

Filtrage des Modèles ML :

Lors de l'exécution "Run All Models", seuls les modèles correspondant aux objectifs accessibles sont exécutés :

```
def run_accessible_models(user, client):
    accessible_objectives = user.userprofile.objectives_access

    results = {}
    if 1 in accessible_objectives:
        results['enrollment'] = run_enrollment_models(client)
    if 2 in accessible_objectives:
        results['churn'] = run_churn_models(client)
    # ... etc pour objectifs 3-6

    return results
```

4.3.6 Système de Chatbot IA

4.3.6.1 Architecture du Chatbot

Le chatbot conversationnel représente l'élément le plus innovant de la plateforme, permettant aux décideurs d'interroger les données et modèles via langage naturel.

Composants Techniques :

- **API IA** : Groq API avec modèle llama-3.3-70b-versatile
- **Backend** : Django REST Framework endpoints
- **Frontend** : Widget JavaScript personnalisé avec WebSocket (polling AJAX)
- **Contexte** : Accès direct à la base de données Client et MLPrediction

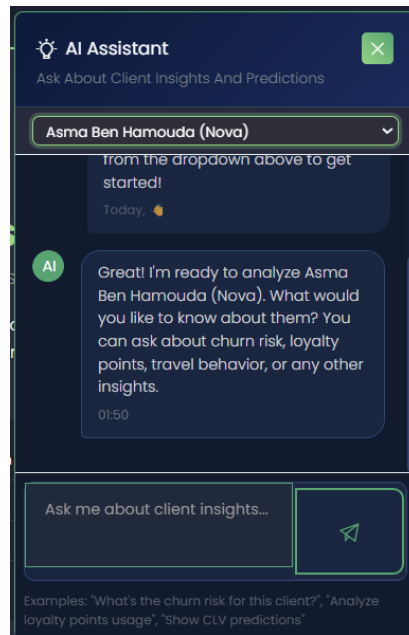


Figure 4.9: Système de Chatbot IA

4.3.6.2 Interface Utilisateur du Chatbot

Widget Permanent :

- **Position** : Icône fixe en bas à droite de l'écran (z-index élevé)
- **Indicateur de disponibilité** : Badge vert "Online" avec animation pulse
- **Ouverture** : Clic sur l'icône déploie la fenêtre de chat (350px × 500px)

Fenêtre de Chat :

- **En-tête** : Titre "Assistant IA" avec bouton de fermeture (×) et indicateur de statut
- **Historique des messages** : Zone de défilement affichant les échanges utilisateur-bot, avec différenciation visuelle (bulles bleues pour l'utilisateur, grises pour le bot)
- **Champ de saisie** : Zone de texte multiligne avec placeholder "Posez votre question sur les clients ou prédictions ML..."
- **Bouton d'envoi** : Bouton "Envoyer" activé uniquement si le champ n'est pas vide, avec icône d'avion pour cohérence thématique
- **Indicateur de frappe** : Animation "IA en train d'écrire..." pendant le traitement de la requête
- **Fonctionnalités avancées** : Boutons rapides pour questions prédéfinies (ex. : "Analyser le churn", "Prédire CLV"), historique des conversations sauvegardé

4.3.6.3 Fonctionnalités du Chatbot

Traitement des Requêtes :

- **Analyse NLP** : Utilisation de l'API Groq pour comprendre les intentions et extraire les entités clés (noms de clients, métriques ML, périodes temporelles)
- **Accès aux Données** : Requêtes SQL dynamiques sur les modèles Client et MLPrediction via Django ORM
- **Génération de Réponses** : Synthèse des résultats en langage naturel, incluant tableaux, graphiques simples ou liens vers vues détaillées

Capabilités Spécifiques :

- **Interrogation de Clients** : Recherche par nom, email, ou critères démographiques avec affichage des prédictions ML associées
- **Analyses Prédictives** : Questions sur le churn, CLV, segmentation, avec explications des modèles utilisés
- **Visualisations** : Génération de graphiques intégrés (via Chart.js) pour tendances temporelles ou distributions
- **Conseils Actionnables** : Suggestions basées sur les prédictions (ex. : "Ce client risque de cherner dans 3 mois, proposez une promotion")

4.3.6.4 Sécurité et Performance

Mesures de Sécurité :

- **Authentification** : Accès réservé aux utilisateurs connectés avec rôles appropriés
- **Sanitisation** : Nettoyage des entrées utilisateur pour prévenir les injections SQL ou XSS
- **Limitation de Taux** : Contrôle du nombre de requêtes par minute pour éviter la surcharge

Optimisation Performance :

- **Caching** : Mise en cache des requêtes fréquentes et des prédictions ML
- **Async Processing** : Traitement asynchrone des requêtes complexes via Celery
- **WebSocket Fallback** : Utilisation de polling AJAX si WebSocket n'est pas disponible

4.4 Conclusion

La plateforme *Airlytics* incarne une approche moderne de l'analytique décisionnelle, où la visualisation, la prédiction et l'interaction convergent pour soutenir une gestion proactive du programme de fidélité. À travers six dashboards stratégiques et une application web intégrée, elle permet non seulement de diagnostiquer l'état actuel de la base client, mais aussi d'anticiper ses évolutions et d'orienter les décisions vers les actions à plus fort impact.

L'architecture modulaire en Django assure une maintenance aisée, une extensibilité vers de nouveaux modèles ML, et une sécurité fine grâce à un système de rôles aligné sur les besoins métier. L'intégration du chatbot IA enrichit cette vision en démocratisant l'accès aux insights : tout utilisateur peut désormais interroger les données en langage naturel, sans compétence technique préalable. Cette solution ne se limite pas à un outil d'analyse ; elle constitue un véritable levier opérationnel pour augmenter la rétention, stimuler la rédemption des points, accélérer la progression des tiers et personnaliser l'expérience client. Elle est ainsi prête à accompagner l'évolution stratégique du programme de fidélité, tout en servant de base à de futures extensions — telles que l'automatisation des campagnes, l'optimisation en temps réel ou l'intégration de nouvelles sources de données.

Conclusion Générale

Le projet *Airlytics* démontre comment une approche **data-driven**, rigoureusement structurée et technologiquement intégrée, peut transformer un programme de fidélité aérien en un moteur d'**équité**, de **croissance** et de **valeur partagée**. Alors que l'introduction soulignait le défi central — exploiter les données client non seulement pour maximiser la rentabilité, mais aussi pour renforcer l'engagement de manière inclusive — cette solution y répond par une architecture unifiée où la **Business Intelligence**, le **Machine Learning** et l'**intelligence conversationnelle** convergent au service de l'action.

En partant d'un entrepôt de données bien structuré (schéma en étoile, ETL robuste) et d'un pipeline de préparation rigoureux (gestion des déséquilibres, encodage stratégique, feature engineering), *Airlytics* déploie six modèles prédictifs alignés sur des objectifs stratégiques concrets : **réduire le churn**, **stimuler la rédemption des points**, **accélérer la progression des tiers**, **personnaliser l'expérience voyageur**, **optimiser la valeur client** et **élargir l'accès aux segments sous-représentés**. Chacun de ces modèles repose sur des métriques de performance solides ($AUC > 0,90$, $R^2 > 0,80$, précision élevée) et s'inscrit dans une logique d'**impact mesurable**.

La plateforme web développée en Django matérialise cette vision en rendant les insights **accessibles**, **actionnables** et **sécurisés**. Grâce à des tableaux de bord Power BI synchronisés, une interface de gestion client intuitive, et un chatbot IA permettant d'interroger les données en langage naturel, *Airlytics* **démocratise l'analytique avancée** — sans sacrifier la rigueur technique ni la sécurité.

Au-delà des performances algorithmiques, ce projet incarne une **philosophie analytique équilibrée** : optimiser la valeur tout en garantissant l'équité, cibler les clients à haut potentiel sans négliger les segments émergents, et automatiser les décisions sans opacité. En cela, *Airlytics* ne se contente pas de répondre aux enjeux actuels de l'industrie aérienne ; il propose un modèle reproductible pour toute organisation cherchant à transformer ses données en levier de **croissance durable et inclusive**.

En somme, là où l'introduction posait la question — « *Comment utiliser les données pour renforcer la fidélité et l'équité simultanément ?* » — la conclusion apporte une réponse claire, opérationnelle et scalable : **en intégrant vision stratégique, rigueur technique et accessibilité décisionnelle au cœur d'une plateforme unifiée**.

Bibliography